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Master thesis

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Ethnic Inequalities in Mental Well-Being at the Onset of COVID-19: Evidence from the UK

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Abstract

This paper examines whether ethnic minority groups in the United Kingdom experienced larger declines in mental well-being at the onset of the COVID-19 pandemic. Using data from Understanding Society and its COVID-19 Study, I compare the White majority (WM) with both a broad Black, Asian, and other minority ethnic (BAME) category and a narrower group of Bangladeshi, Indian, and Pakistani (BIP) respondents. Empirically, I combine Oaxaca–Blinder decompositions of changes in GHQ-12 mental well-being between the 2019 pre-pandemic baseline and April 2020 with an event-study approach that traces the evolution of ethnic mental well-being gaps before and after the onset of the pandemic. The results show that the broad BAME–White majority comparison does not provide strong evidence of a large or persistent widening of the mental well-being gap. However, a larger and statistically significant gap emerges when the minority group is restricted to Bangladeshi, Indian, and Pakistani respondents. The decomposition results suggest that age composition and economic vulnerability, particularly employment and financial conditions, are important factors associated with this gap. The event-study results further show a sharp widening of the BIP–White majority gap in the early months of the pandemic, followed by partial but incomplete convergence. These findings highlight the importance of ethnic categorization in studies of health inequality and suggest that the deterioration in mental well-being associated with the onset of COVID-19 was more pronounced among specific minority groups facing greater socioeconomic vulnerability.

CONTENTS

1 Introduction

The COVID-19 pandemic generated a major shock to mental well-being in the United Kingdom. Beyond the direct health risks of infection, the pandemic and associated lockdown measures disrupted employment, income security, household routines, and social interaction. Existing evidence shows that mental health deteriorated substantially during the early stages of the pandemic (Li and Wang 2020; Daly, Sutin, and Robinson 2022). This deterioration, however, was unlikely to be evenly distributed across the population. Individuals and households entered the pandemic with different levels of economic security, housing conditions, health risks, and caring responsibilities. Women and young adults, for example, were more likely to experience psychological distress during the pandemic (Li and Wang 2020; Niedzwiedz et al. 2021; Banks and Xu 2020), partly reflecting young adults' weaker labour-market attachment (Cortes and Forsythe 2023) and women's greater unpaid care responsibilities during lockdown (Xue and Anne McMunn 2021). These differences may have shaped both individuals' exposure to the pandemic shock and their capacity to absorb its consequences.

Ethnic minority groups may have been particularly vulnerable to these disruptions. In the UK, minority ethnic groups differ from the White majority[1] in several dimensions that are closely related to mental well-being, including labour market position, financial resilience, housing conditions, pre-existing health risks, household composition, and demographic structure. For example, some minority groups were more concentrated in shutdown sectors or key-worker occupations, while others were more likely to live in larger households or to have weaker financial buffers before the pandemic (Platt and Warwick 2020). These differences suggest that the pandemic may have widened existing ethnic inequalities in mental health. At the same time, ethnic minority groups are highly heterogeneous. Broad categories such as BAME may combine groups with very different social contexts and pandemic experiences, potentially obscuring the groups most affected by the crisis.

This paper examines whether ethnic minority groups experienced larger declines in mental well-being than the White majority at the onset of the COVID-19 pandemic in the UK. The analysis

[1] British/English/Scottish/Welsh/Northern Irish.

focuses on changes in mental well-being rather than levels, in order to examine whether ethnic gaps widened after the pandemic shock. I first compare the White majority (WM) with a broad Black, Asian, and other minority ethnic (BAME) category. I then restrict the minority sample to Bangladeshi, Indian, and Pakistani (BIP) respondents, who may have faced particularly high exposure to economic and household-related vulnerabilities during the early stage of the pandemic. This distinction is important because the broad BAME category may mask substantial heterogeneity across minority groups.

The analysis uses data from Understanding Society and the Understanding Society COVID-19 Study. Mental well-being is measured using the GHQ-12 Likert score, which is standardized, inverted, and seasonally adjusted so that higher values indicate better mental well-being. The pre-pandemic baseline is constructed using interviews from the 2019 calendar year, while the first wave of the COVID-19 Study in April 2020 captures the immediate onset of the pandemic and the first national lockdown. This structure allows the paper to examine individual-level changes in mental well-being from before to after the onset of COVID-19.

I use two complementary empirical approaches. The Oaxaca–Blinder decomposition examines whether observed socioeconomic and demographic characteristics account for ethnic differences in changes in mental well-being. It separates the gap into a composition component, reflecting differences in group characteristics, and a structural component, reflecting differences in group-specific associations between these characteristics and changes in mental well-being. The explanatory variables are grouped into six blocks: employment and financial conditions, housing conditions, health, household composition, sex, and age. The event-study analysis then traces the evolution of ethnic mental-health gaps before and after the onset of COVID-19. This approach assesses whether the gaps widened sharply after the pandemic began or instead reflected pre-existing differential trends.

The results show that the broad BAME–White majority comparison does not provide strong evidence of a large or persistent widening of the mental-health gap. Although mental well-being declined among both groups after the onset of the pandemic, the estimated BAME–White majority gap is small and imprecisely estimated. The event-study results also suggest that the broad BAME gap was already evolving before the pandemic, making it difficult to interpret post-pandemic differences as a newly generated gap.

The results differ when the minority group is restricted to Bangladeshi, Indian, and Pakistani respondents. The WM–BIP gap in mental-health changes is larger and statistically significant. The decomposition results suggest that age composition and economic vulnerability, especially employment and financial conditions, are important factors associated with this gap. Moreover, the structural component for employment and financial conditions is substantial, suggesting that similar economic vulnerabilities were associated with larger declines in mental well-being among BIP respondents than among the White majority. The event-study results reinforce this interpretation: unlike the broad BAME comparison, there is no clear evidence of differential pre-pandemic trends for BIP respondents, while the BIP–WM gap widens sharply in the first months of the pandemic and only partially narrows later.

This paper contributes to the literature in two main ways. First, it highlights the importance of ethnic categorization when studying health inequalities. While broad minority categories can be useful for retaining smaller groups in the analysis, they may also obscure substantial heterogeneity in pandemic experiences. Second, the paper provides evidence that economic vulnerability played an important role in the widening of mental-health inequalities among Bangladeshi, Indian, and Pakistani respondents at the onset of COVID-19. The findings suggest that ethnic inequalities in mental health during the pandemic cannot be understood only through healthcare access or infection risk, but should also be examined in relation to employment insecurity, financial hardship, and broader socio-economic vulnerability.

The remainder of the paper is organized as follows. Section 2 describes the data and discusses the potential mechanisms through which minority ethnic groups may have been more vulnerable to the pandemic shock. Section 3 presents the empirical strategy, including the Oaxaca–Blinder decomposition and the event-study design. Section 4 reports the empirical results. Section 5 discusses the interpretation, limitations, and implications of the findings. Section 6 is the conclusion.

2 Data and Conceptual Framework

Data for this paper are drawn from Understanding Society (University of Essex, Institute for Social and Economic Research 2025) and its COVID-19 Study (University of Essex, Institute for Social and Economic Research 2021), a large longitudinal household panel survey in the United Kingdom. The surveys provide information on respondents' ethnic background, mental well-being, and socioeconomic and demographic characteristics. I construct a pre-pandemic baseline using interviews conducted in the 2019 calendar year, corresponding to the second year of Wave 10 and the first year of Wave 11 of the main survey. The first wave of the COVID-19 Study, conducted in April 2020, is used to capture the immediate period following the onset of the pandemic and the first national lockdown. A total of 16,152 respondents were interviewed in the 2019 baseline year, of whom 14,793 were re-interviewed in the COVID-19 Study wave used in the analysis. For decomposition analyses based on the 2019–April 2020 sample, I use COVID-19 cross-sectional weights[2] and adjust standard errors for the survey design.[3] The event-study analysis is estimated using an unweighted individual fixed-effects specification, as discussed in Section 3.

Respondents' ethnicity is measured using *ethn_dv* from *Understanding Society*. Since the survey collects a wide range of ethnicity-related information, including ethnic group, national identity, country of birth, parental and grandparental country of birth, and childhood language, this variable is treated as the survey's standardized derived measure of ethnicity. The White majority (WM) is defined as respondents who identify as British, English, Scottish, Welsh, or Northern Irish. I first compare this group with a broad Black, Asian, and other Minority Ethnic (BAME) category, defined as all respondents outside the White-majority group.[4] Although this aggregation may obscure heterogeneity across specific minority groups, it allows the inclusion of underrepresented ethnic groups in the analysis. I then restrict the minority ethnic sample to Bangladeshi, Indian, and Pakistani (BIP) respondents, a group for which existing evidence and descriptive patterns suggest

[2] These weights are based on the Wave 10 cross-sectional analysis weights and are further adjusted for non-response in each wave of the COVID-19 survey.

[3] The survey design adjustment accounts for primary sampling units and strata.

[4] This broad grouping is used as a pragmatic aggregate comparison and may include respondents from other White backgrounds; it should therefore be interpreted as a broad minority comparison rather than as a strictly non-White classification.

greater exposure to economic and household-related vulnerabilities at the onset of the pandemic.

My focal variable, mental well-being, is measured using the 12-item General Health Questionnaire (GHQ-12). The GHQ is a widely used self-reported instrument for measuring non-psychotic psychological distress. Because the analysis focuses on continuous variation in mental well-being rather than on identifying probable psychiatric cases using a threshold, I use the Likert scoring method, which codes each item on a 0-1-2-3 scale, rather than the caseness method, which collapses responses into a 0-0-1-1 format. The original GHQ-12 Likert score ranges from 0 to 36, with higher values indicating greater psychological distress. I invert the score so that higher values correspond to better mental well-being. The resulting measure is then standardized and seasonally adjusted. Figure 1 provides an initial descriptive overview of mental well-being trends before and after the onset of COVID-19. Both BAME and White-majority respondents experienced a sharp decline after the pandemic began, with the decline appearing somewhat larger among BAME respondents in the early COVID-19 waves. Since the figure is descriptive and uses the broad BAME category, it should be interpreted as motivation for the subsequent analysis rather than as evidence of a causal or persistent widening of the ethnic gap.

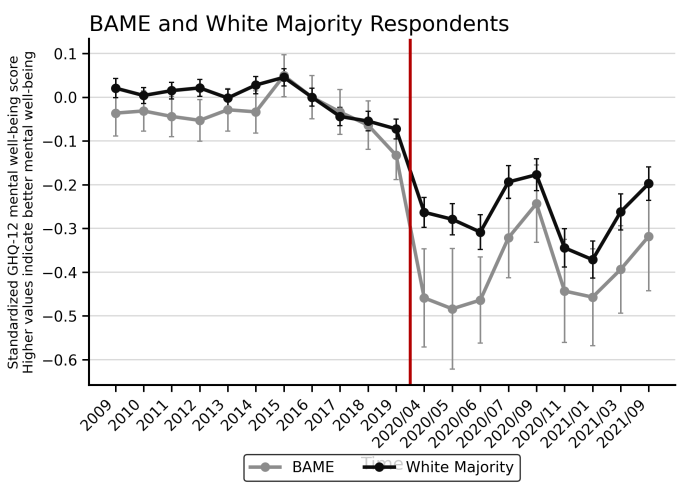

Figure 1: Descriptive Trends in Mental Well-Being among BAME and White Majority Respondents

Note: Data are drawn from Understanding Society main survey waves 1–11 and the Understanding Society COVID-19 Study. The figure reports group-specific weighted means of the seasonally adjusted, inverted, and 2019-standardized GHQ-12 Likert score. Higher values indicate better mental well-being. Means are indexed to zero in 2016 within each group. Main-survey observations are grouped by calendar year of interview, while COVID-19 Study observations are grouped by survey wave. Self-completion cross-sectional weights are used for the main survey, and wave-specific cross-sectional respondent weights are used for the COVID-19 Study. Error bars indicate approximate 95% confidence intervals.

Table 1 complements this aggregate trend by reporting ethnicity-specific changes in mental well-being between the 2019 baseline and April 2020. The table shows substantial heterogeneity across minority groups. Respondents from "Any other mixed background" experienced the largest deterioration, while Pakistani, Indian, and Bangladeshi respondents also experienced larger average declines than the White British group. By contrast, some other minority groups show smaller declines or even improvements. This heterogeneity motivates the distinction between the broad BAME comparison and the narrower BIP analysis used below.

Table 1: Changes in GHQ-12 Mental Well-Being by Ethnic Group, 2019–April 2020

Ethnicity	2019 baseline	April 2020	Change	SD of change	N
Any other mixed background	-0.10	-0.93	-0.83	1.13	62
Any other black background	0.56	-0.25	-0.81	0.95	13
Arab	0.10	-0.46	-0.55	1.26	31
Pakistani	0.20	-0.21	-0.41	1.35	265
Any other White background	0.10	-0.29	-0.39	1.03	419
Indian	0.04	-0.28	-0.32	0.98	435
Chinese	0.11	-0.18	-0.29	0.95	63
Bangladeshi	-0.25	-0.52	-0.27	1.45	101
White and Black Caribbean	-0.59	-0.83	-0.24	1.04	90
African	-0.29	-0.49	-0.20	1.11	114
British/English/Scot-tish/Welsh/Northern Irish	0.03	-0.15	-0.18	0.99	12,581
Any other Asian background	0.09	0.01	-0.09	1.03	96
White and Black African	0.05	-0.03	-0.08	0.84	32
Irish	-0.06	-0.14	-0.07	1.12	249
White and Asian	0.12	0.07	-0.05	1.20	87
Any other ethnic group	-0.82	-0.56	0.26	1.18	24
Caribbean	-0.55	-0.17	0.38	1.03	131

Notes: Data are drawn from the 2019 UKHLS main survey and the April 2020 Understanding Society COVID-19 Study. The table reports ethnicity-specific descriptive statistics for the April-adjusted, standardized, and inverted GHQ-12 Likert score. Higher values indicate better mental well-being. The 2019 baseline and April 2020 columns report weighted means. Change is defined as the individual-level difference between April 2020 and 2019. SD denotes the unweighted standard deviation of the change, and N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

A potential concern is that ethnic differences in GHQ outcomes may partly reflect differential reporting styles rather than true differences in mental distress. However, King et al. (2023) find no clear evidence that the GHQ-12 exhibits ethnic-group bias in population-level comparisons in the UK. This suggests that ethnic gaps in GHQ outcomes are unlikely to be driven solely by differential reporting styles.

Potential Mechanisms Underlying Ethnic Differences in Mental Well-Being

Ethnic differences in pandemic-related risks should be interpreted in relation to the social and economic conditions that shaped exposure to infection risk, economic disruption, and household stress, rather than as differences intrinsic to ethnic groups. For example, occupational sorting into health care, transport, frontline work, or shutdown sectors may have increased some groups' exposure to the health and economic consequences of the pandemic. Interpreting ethnic gaps in this context is important for avoiding explanations that reinforce entrenched stereotypes or overlook structural sources of inequality (Asaria et al. 2020; Platt and Warwick 2020). This section reviews socioeconomic and demographic conditions that may have shaped ethnic differences in exposure to the pandemic shock and in subsequent changes in mental well-being. The corresponding variables used in the decomposition analysis are introduced at the end of each subsection.

Labour Market Disruption and Employment Status

Labour-market disruption was one of the main channels through which the COVID-19 pandemic may have affected mental well-being (Posel, Oyenubi, and Kollamparambil 2021; Guerin et al. 2021; Wörn, Reme, and Skirbekk 2023; Ruengorn et al. 2021; Li and Wang 2020). In the UK, the immediate rise in measured unemployment was partly attenuated by policy support, especially the Coronavirus Job Retention Scheme (CJRS), which allowed many workers to remain formally attached to their employers, and the Self-Employment Income Support Scheme (SEISS), which provided support to eligible self-employed workers. However, these schemes did not eliminate employment and income insecurity. Workers in shut-down sectors, the self-employed, and those experiencing furlough or reduced working hours still faced substantial uncertainty, which may have contributed to deterioration in mental well-being.

Even with these protections, labour-market risks were unevenly distributed across ethnic groups. Workers in shut-down sectors and the self-employed were more likely to experience earnings losses during lockdown, and these groups were not evenly represented across ethnic groups (Blundell et al. 2020; Platt and Warwick 2020). Bangladeshi and Pakistani men were particularly concentrated in shut-down industries such as passenger transport and food services, while Black African and Black Caribbean men were also more likely than White British men to work in such sectors. Self-employed workers also faced uncertainty because SEISS payments were delayed, and some workers were either ineligible or received support below their previous earnings. Key workers faced a different type of risk. They were less exposed to job loss, but more exposed to infection risk, heavier workloads, and role strain because they continued to work in essential sectors (Blundell et al. 2020; Sze et al. 2020; Pan et al. 2020). Pakistani, Black, and Indian men were disproportionately represented among key workers (Platt and Warwick 2020). The expected relationship between key worker status and mental well-being is therefore ambiguous. Qadri (2026) finds that key workers experienced larger increases in psychological distress during the pandemic, especially non-health-and-social-care key workers, while Serra-Sastre, Pinilla, and Kalansooriya (2026) find no significantly worse mental-health trajectories among health and social care workers. These mixed findings suggest that key worker status may capture both adverse occupational exposure and potentially protective factors, such as continued employment, income stability, and social contact.[5] I therefore include key worker status separately within the employment and financial conditions block, since it may capture both adverse occupational exposure and potentially protective employment stability.

To capture labour-market disruption during COVID-19, I classify respondents who were employed before the pandemic into three categories: no change, reduction, and job loss. Respondents who were furloughed or experienced reduced working hours are classified as experiencing a reduction, while those who lost their jobs are classified as experiencing job loss. I also identify key workers who did not experience employment disruption, in order to capture exposure to continued frontline work separately from job loss or reduced hours. Table 2 shows substantial ethnic differences in employment disruption and work status among respondents who were employed before the pandemic. Descriptively, job loss was highest among Bangladeshi respondents, followed by African and Pakistani respondents, although these estimates should be interpreted cautiously because of small unweighted sample sizes. Reductions in working hours or furlough were espe-

[5] Differences in the pandemic periods examined may also help explain these divergent findings. Qadri (2026) estimates mental health changes at three infection peaks—May 2020, November 2020, and January 2021—whereas Serra-Sastre, Pinilla, and Kalansooriya (2026) pool April to November 2020 into a single short-term post-pandemic period.

cially common among Pakistani respondents and respondents from “Any other White background”. Indian and Bangladeshi respondents were more likely to remain in work as key workers, while respondents from “Any other White background” had the highest rate of self-employment.[6] These patterns suggest that ethnic groups were exposed to different types of labour-market risk during the early stage of the pandemic: some faced greater risk of job loss or reduced hours, while others faced continued frontline work and associated health risks.

Table 2: Employment Disruption and Work Status among Pre-Pandemic Workers by Ethnic Group

Ethnicity	Reduction	Job loss	Key worker	Self-employed	N
British/English/Scottish/Welsh/Northern Irish	46.77	4.06	29.25	11.55	7,939
Indian	30.91	6.87	37.00	12.55	135
Pakistani	50.50	8.52	24.02	9.37	81
Bangladeshi	34.63	18.88	39.36	8.30	37
African	48.10	9.47	29.11	1.75	51
Caribbean	46.14	2.70	24.46	4.32	62
Any other white background	50.80	3.78	19.07	21.32	274

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The sample is restricted to respondents who were employed before the pandemic. The table reports weighted percentages by ethnic group. “Reduced hours or furlough” includes respondents who were furloughed or experienced a reduction in working hours during the COVID-19 wave. “Job loss” indicates respondents who were no longer employed in April 2020. “Key worker, no disruption” identifies key workers who reported no change in employment status. “Self-employed only” refers to respondents who reported being self-employed only before the pandemic, excluding those who were employees only or both employees and self-employed. N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

Financial Difficulties and Economic Vulnerability

Financial vulnerability is another channel through which the pandemic may have affected mental well-being. COVID-related financial constraints, including income loss and financial stress,

[6] This category includes only respondents who reported being self-employed only, excluding those who were employed only or both employed and self-employed.

have been associated with poorer mental health outcomes (Richardson et al. 2025). Lower savings and higher debt may also increase financial stress, which in turn predicts declines in mental health (Simonse et al. 2022). These risks were not evenly distributed across ethnic groups. During COVID-19, only around 30% of Bangladeshi, Black Caribbean, and Black African households were able to cover one month of income, compared with nearly 60% among other groups (Platt and Warwick 2020). This weaker financial buffer partly reflected lower access to multiple earners within the household and lower partner earnings where a partner was employed. It was also compounded by greater concentration in shut-down sectors, suggesting that financial vulnerability should be understood as part of a broader pattern of structural economic disadvantage rather than as an isolated explanation.

I distinguish between pre-existing financial hardship and changes in financial status during the pandemic. The former captures baseline economic vulnerability, while the latter captures deterioration in financial circumstances after the onset of COVID-19. Baseline financial difficulties are defined using respondents' self-assessed financial circumstances[7] in 2019. Respondents who reported that they were "just about getting by," "finding it quite difficult," or "finding it very difficult" are classified as experiencing financial difficulties. Table 3 shows substantial ethnic differences in both pre-existing financial hardship and pandemic-related financial deterioration. African and Caribbean respondents reported the highest levels of baseline financial difficulties, while Bangladeshi respondents also faced relatively high pre-pandemic hardship. By contrast, the White British group reported substantially lower baseline financial difficulties. The pattern differs somewhat for changes during the pandemic: Bangladeshi, Pakistani, and Indian respondents showed the highest shares of worsened financial circumstances. These descriptive patterns suggest that financial vulnerability operated through both pre-existing hardship and deterioration after the onset of COVID-19, with different minority groups exposed to these dimensions in different ways.

[7]Respondents were asked: "How well would you say you yourself are managing financially these days? Would you say you are..."

Table 3: Pre-Pandemic Financial Difficulties and Financial Deterioration by Ethnic Group

Ethnicity	Baseline difficulties	Worsened financial situation	N
British/English/Scottish/Welsh/Northern Irish	26.60	15.96	12,558
Indian	33.57	21.81	433
Pakistani	26.18	23.71	262
Bangladeshi	47.52	24.03	100
African	66.07	14.54	112
Caribbean	62.56	17.77	131
Any other white background	31.35	19.46	417

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports weighted percentages by ethnic group. "Baseline financial difficulties" identifies respondents who reported in 2019 that they were just about getting by, finding it quite difficult, or finding it very difficult financially. "Worsened financial situation" identifies respondents whose financial situation in April 2020 was worse than before the pandemic. Percentages are weighted using the COVID-19 survey weights, and N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

Housing Conditions and Residential Crowding

Ethnic disparities in housing conditions are well documented in England and Wales (Coulter 2025). Some minority groups, including Arab (25%), Bangladeshi (18%), and Black African households (16%), are more likely to live in overcrowded accommodation than White British households (2%) (Ministry of Housing, Communities and Local Government 2025). These disparities should not be interpreted simply as the result of household preferences or family size. Evidence on South Asian households (Department for Levelling Up, Housing and Communities 2022) suggests that overcrowding reflects a combination of structural constraints, including limited access to suitable and affordable housing, long waiting times for social housing, and the importance of local family and support networks. Housing constraints may therefore represent another channel through which pre-existing socioeconomic disadvantage shaped exposure to the pandemic shock.

Housing space may matter for mental well-being because limited space can reduce privacy, increase household stress, and constrain the ability to work, study, rest, or care for family members at home (Amerio et al. 2020; Mangrio and Zdravkovic 2018; Groot et al. 2022; Pengcheng et al. 2021; Evans 2003). These mechanisms may have become more important during lockdown, when individuals spent more time inside the home (Amerio et al. 2020). Existing studies generally find that crowding and limited housing space are associated with poorer mental health outcomes, although the evidence also suggests that these associations are closely related to broader socioeconomic circumstances, housing stability, income, and family structure. For example, Mangrio and Zdravkovic (2018) find no significant association between crowding and mental health after controlling for housing stability, while Pengcheng et al. (2021) find that the association is concentrated among low-income households. This suggests that housing conditions should be interpreted cautiously: the measures used here capture space constraints, but not all dimensions of housing quality, such as neighborhood conditions, access to green space (Akbari et al. 2021), or housing stability.

Table 4 reports two measures of pre-pandemic housing space: the bedroom-per-person ratio and the number of other rooms in the accommodation, excluding kitchens and bathrooms. These measures capture different dimensions of residential space. The bedroom-per-person ratio reflects sleeping-space constraints, while the number of other rooms captures broader living space available within the household. Pakistani and Bangladeshi respondents were especially likely to live in households with fewer bedrooms than residents, and very few had two or more bedrooms per person. Indian respondents also faced more limited bedroom space than the White British group. The distribution of other rooms shows a somewhat different pattern: Bangladeshi, African, and Caribbean respondents were more concentrated in homes with only 0–1 other rooms. These patterns suggest that housing constraints may have been one channel through which the pandemic affected mental well-being, although the estimates for smaller groups should be interpreted cautiously.

Table 4: Pre-Pandemic Housing Space by Ethnic Group

	Bedrooms per person			Other rooms in accommodation			
Ethnicity	<1	1–<2	≥2	0–1	2–3	4+	N
British/English/Scottish/Welsh/Northern Irish	22.91	54.70	22.38	39.52	52.25	8.23	12,385
Indian	46.87	45.80	7.33	37.76	56.40	5.84	417
Pakistani	76.40	20.30	3.30	48.19	48.45	3.36	250
Bangladeshi	78.71	20.64	0.65	74.50	23.37	2.13	92
African	35.30	61.13	3.56	63.06	35.48	1.45	107
Caribbean	19.62	58.71	21.66	65.77	31.33	2.90	129
Any other white background	29.01	55.78	15.21	48.94	44.08	6.98	406

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports weighted percentages of pre-pandemic housing space by ethnic group. The first three columns show the distribution of the bedroom-per-person ratio, while the next three columns show the distribution of the number of other rooms in the accommodation, excluding kitchens and bathrooms. Percentages are weighted using the COVID-19 survey weights, and N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

Pre-Existing Health Conditions and COVID-19 Symptoms

Pre-existing health conditions may have shaped vulnerability to the pandemic shock. Some ethnic minority groups in the UK, particularly South Asian and Black groups, face a higher burden of multiple long-term conditions than the White British population (Hayanga, Stafford, and Bécares 2023). These differences should not be interpreted as intrinsic biological or cultural differences across ethnic groups. Rather, existing evidence links ethnic inequalities in long-term conditions to structural disadvantage, including unequal exposure to socioeconomic risks, discrimination, and differences in access to health-relevant resources.

Whether pre-existing health conditions translated into larger deterioration in mental well-being during COVID-19 is less clear. Buneviciene et al. (2022) find that pre-existing health conditions are associated with poorer mental health, while Budu et al. (2021) find no significant association. More recent evidence Rünger et al. (2025) suggests that poorer mental health among individuals with chronic conditions may reflect worse baseline mental health rather than larger pandemic-induced deterioration. I therefore include chronic conditions as part of the health-related vulnerability block, rather than interpreting them as capturing the causal effect of pre-existing physical health conditions on changes in mental well-being.

I measure chronic conditions using the long-term health condition module in the April 2020 COVID-19 Study. Respondents were asked whether a doctor or other health professional had ever told them that they had specific conditions. I group these conditions into respiratory,[8] cardiovascular,[9] endocrine,[10] arthritis, and other chronic conditions.[11] I also include reported COVID-19 symptoms, as such symptoms may directly capture exposure to the health risks of the pandemic (Li and Wang 2020). Table 5 shows that African respondents reported the highest share of COVID-19 symptoms and a relatively high prevalence of other chronic conditions, while Bangladeshi respondents also reported a relatively high rate of COVID-19 symptoms. Caribbean respondents reported particularly high rates of arthritis and cardiovascular conditions. Pakistani and Indian respondents reported lower prevalence of several diagnosed chronic conditions, although these differences should be interpreted cautiously given possible under-diagnosis, differences in healthcare access, and small sample sizes for some groups.

[8] Asthma, emphysema, chronic bronchitis, and COPD (Chronic Obstructive Pulmonary Disease).

[9] Congestive heart failure, coronary heart disease, angina, heart attack or myocardial infarction, stroke, and high blood pressure/hypertension.

[10] Hypothyroidism or an under-active thyroid and diabetes.

[11] Any kind of liver condition, cancer or malignancy, epilepsy, an emotional, nervous or psychiatric problem, multiple sclerosis, chronic kidney disease, conditions affecting the brain and nerves such as Parkinson's disease, motor neurone disease, a learning disability or cerebral palsy, severe obesity (BMI of 40 or above), and other long-standing/chronic conditions.

Table 5: COVID-19 Symptoms and Chronic Conditions by Ethnic Group

	COVID-19	Chronic conditions					
Ethnicity	Symptoms	Respiratory	Cardio-vascular	Endocrine	Arthritis	Other	N
British/English/Scottish/Welsh/Northern Irish	11.38	16.03	20.56	10.07	12.37	21.93	12,534
Indian	13.92	8.94	8.34	18.94	6.24	13.90	433
Pakistani	9.12	10.25	4.40	6.20	3.94	4.90	265
Bangladeshi	19.46	7.72	9.35	11.94	1.02	3.58	100
African	30.23	9.14	16.64	8.10	6.54	32.46	112
Caribbean	8.58	16.95	26.08	15.43	36.60	27.01	131
Any other white background	17.87	11.35	13.17	9.23	4.20	21.99	418

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports weighted percentages of reported COVID-19 symptoms and selected chronic conditions by ethnic group. COVID-19 symptoms are measured in the April 2020 COVID-19 Study. Chronic conditions are based on whether respondents reported that a doctor or other health professional had ever told them that they had the relevant condition. Respiratory, cardiovascular, endocrine, arthritis, and other refer to the corresponding chronic-condition categories. Percentages are weighted using the COVID-19 survey weights, and N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

Household Composition and Care Responsibilities

Household composition may have shaped exposure to the economic and psychological consequences of the COVID-19 pandemic (Platt and Warwick 2020). Households with dependent children were more exposed to school and childcare disruptions, while single-adult households had fewer opportunities to share care responsibilities or buffer employment and income shocks within the household. These risks were not evenly distributed across ethnic groups. Pakistani and Bangladeshi households were more likely to have dependent children (Kulu and Hannemann 2016) and to live in single-earner households (Khoudja and Platt 2018; Dale et al. 2002; Aston et al. 2007), while Black African and Black Caribbean groups had relatively high shares of lone parents with dependent children (Kiernan and Mensah 2010; Mikolai and Kulu 2023). These differences suggest that household composition may have affected both exposure to pandemic-related disruptions and the capacity to absorb them.

The relationship between household composition and mental well-being is theoretically ambiguous. Dependent children may increase psychological distress through childcare and home-schooling burdens, especially during school closures and in households where care responsibilities cannot be shared (Blanden et al. 2021; Michael J Green et al. 2023). At the same time, co-residence with other adults may reduce loneliness and provide emotional or practical support (Li and Wang 2020). Empirical studies similarly find heterogeneous effects by parental status, the number and age of children, gender, employment status, and the stage of the pandemic (Smail et al. 2022; Chen, McMunn, and Gagné 2023). This motivates including household composition as a separate block in the decomposition rather than assuming a uniform effect of children or household size.

To keep the analysis parsimonious, I use household structure and the number of dependent children measured at the pre-pandemic baseline. I do not incorporate the age of dependent children, which may also be relevant for childcare intensity. Table 6 shows substantial descriptive differences in household composition across ethnic groups. African and Caribbean respondents were more likely to live in single-adult households with dependent children, while Bangladeshi and Pakistani respondents were more likely to live with three or more dependent children. These patterns suggest that pandemic-related care burdens and intra-household support may have differed across ethnic groups.

Table 6: Pre-Pandemic Household Composition by Ethnic Group

	Household structure		Dependent children			
Ethnicity	Single adult	Single adult with dependent children	No	1–2	3+	N
British/English/Scottish/Welsh/Northern Irish	4.69	3.03	71.01	25.25	3.74	10,910
Indian	0.67	0.66	58.33	38.52	3.15	409
Pakistani	4.98	0.91	67.92	17.62	14.46	257
Bangladeshi	0.86	0.26	48.10	30.86	21.05	98
African	7.53	6.98	53.87	35.81	10.32	93
Caribbean	7.75	6.09	68.06	30.01	1.93	102
Any other white background	4.25	3.98	53.29	43.83	2.88	365

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports weighted percentages of pre-pandemic household composition by ethnic group. “Single” identifies respondents living in a single-adult household, while “single adult with dependent children” identifies single-adult households with dependent children. The dependent-children columns report the distribution of the number of dependent children at baseline and are mutually exclusive. Percentages are weighted using the COVID-19 survey weights, and N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

Sex and Age Composition

Sex and age are important predictors of mental well-being during the COVID-19 pandemic. Several studies show that women and younger adults experienced larger increases in psychological distress in the UK during the pandemic (Li and Wang 2020; Niedzwiedz et al. 2021; Banks and Xu 2020; Daly, Sutin, and Robinson 2022; Maddock et al. 2022; Dotsikas et al. 2023; Russell Jonsson et al. 2023). Ethnic differences in mental-health deterioration may therefore partly reflect differences in sex and age composition across groups. At the same time, these characteristics may interact with other pandemic-related vulnerabilities. For example, Proto and Quintana-Domeque (2021) show that men from minority ethnic groups experienced larger increases in psychiatric distress than

White British men, with particularly large deterioration among Bangladeshi, Indian, and Pakistani men. Occupational profiles and household care responsibilities may partly explain these gendered patterns.

Age composition may also matter. Many minority ethnic groups in the UK have younger age profiles than the White British majority, reflecting migration history and demographic processes (Shankley, Hannemann, and Simpson 2020). This is relevant because younger adults were more exposed to age-specific mental-health risks during the pandemic, including labour-market insecurity (Daly, Sutin, and Robinson 2022) and increased loneliness (Niedzwiedz et al. 2021). Although age structure alone cannot explain ethnic inequalities in mental well-being, it may have interacted with other pandemic-related stressors.

For the empirical analysis, I classify age into four categories: 16–29, 30–49, 50–69, and 70 or above. Table 7 reports sex and age composition by ethnic group. Pakistani and Bangladeshi respondents were noticeably younger than the White British group, with larger shares in the 16–29 and 30–49 age groups and very small shares aged 70 or above. Caribbean respondents were relatively older, with more than half in the 50–69 age group, and women made up a particularly large share of this group. These patterns suggest that sex and age composition should be accounted for when comparing changes in mental well-being across ethnic groups.

Table 7: Pre-Pandemic Sex and Age Composition by Ethnic Group

Ethnicity	Female	Age 16–29	Age 30–49	Age 50–69	Age 70+	N
British/English/Scottish/Welsh/Northern Irish	53.53	16.76	27.96	37.70	17.58	12,575
Indian	45.23	27.99	43.71	24.90	3.41	435
Pakistani	42.66	58.10	30.10	11.27	0.53	263
Bangladeshi	54.77	42.92	51.68	3.97	1.44	100
African	42.84	15.20	60.39	17.38	7.03	111
Caribbean	70.18	19.76	25.57	54.21	0.47	131
Any other white background	58.53	8.78	56.01	28.48	6.73	417

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports weighted percentages of pre-pandemic sex and age composition by ethnic group. "Female" indicates the share of women. The remaining columns report the distribution across age groups. Percentages are weighted using the COVID-19 survey weights, and N denotes the unweighted sample size. Estimates for groups with small unweighted sample sizes should be interpreted with caution.

3 Method

Oaxaca-Blinder decomposition

To examine the factors associated with ethnic differences in changes in mental well-being at the onset of the COVID-19 pandemic, I use the Oaxaca–Blinder decomposition approach (Oaxaca 1973; Blinder 1973). Although the method was originally developed to decompose wage differentials, it can also be used to decompose differences in changes in mental well-being between groups. In this application, the outcome is the change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020.[12] Higher values indicate better mental well-being.

Let M denote the minority group in a given comparison, where M is defined either as BAME

[12] Standardization is based on the weighted mean and weighted standard deviation computed using the full sample of April 2020 COVID-module respondents, drawing on all linked observations retained across waves. All pre-pandemic observations are first adjusted to their April-equivalent values, and the sample is only subsequently restricted to respondents observed in both 2019 and the post-onset period.

or as BIP. To keep the sign of the decomposition consistent with the WM–minority gap, I define WM_i as an indicator equal to one if individual i belongs to the White majority and zero if individual i belongs to group M. I begin with the pooled specification:

$$\Delta ghq_i = \alpha + \beta WM_i + \delta_1' \Delta Z_{1,i} + \delta_2' Z_{2,i,2019} + \varepsilon_i. \tag{1}$$

Here, $\Delta Z_{1,i}$ denotes pandemic-period changes, such as employment disruption and changes in financial status, while $Z_{2,i,2019}$ denotes baseline characteristics measured before the pandemic, such as sex, age, housing conditions, chronic conditions, and household composition. The coefficient β captures the conditional WM–minority difference in changes in mental well-being, holding the observed characteristics fixed. Since higher values of the outcome indicate better mental well-being, a positive β indicates that, conditional on observed characteristics, the minority group experienced a larger deterioration in mental well-being than the White majority.

I then allow the associations between the explanatory variables and changes in mental well-being to differ across groups. The group-specific specification is:

$$\Delta ghq_i = \alpha_e + \delta_{e1}' \Delta Z_{1,i} + \delta_{e2}' Z_{2,i,2019} + \varepsilon_i, \tag{2}$$

where $e \in \{WM, M\}$. This specification allows the White majority and the minority group to have different associations between the observed characteristics and changes in mental well-being.

Models (1) and (2) provide the basis for the Oaxaca–Blinder decomposition. Let $\Delta\overline{ghq}^{e}$ denote the mean change in mental well-being for group e. Taking the difference in group means, the WM–minority gap can first be written as:

$$\Delta\overline{ghq}^{WM} - \Delta\overline{ghq}^{M} = \sum_{x=1}^{2} \left[\delta_x' \left(\overline{Z}_x^{WM} - \overline{Z}_x^{M} \right) \right] + \beta. \tag{3}$$

This gap can then be decomposed into a composition effect and a structural effect:

$$\Delta\overline{ghq}^{WM} - \Delta\overline{ghq}^{M} = \underbrace{\sum_{x=1}^{2}\left[\delta'_x\left(\overline{Z}_x^{WM} - \overline{Z}_x^{M}\right)\right]}_{\text{composition effect}}$$
$$+ \underbrace{\left[\sum_{x=1}^{2}\left(\delta'_{WM,x} - \delta'_x\right)\overline{Z}_x^{WM} + \sum_{x=1}^{2}\left(\delta'_x - \delta'_{M,x}\right)\overline{Z}_x^{M} + \alpha_{WM} - \alpha_M\right]}_{\text{structural effect}}. \tag{4}$$

The first term is the composition effect, which captures the part of the WM–minority gap associated with differences in observed characteristics. The remaining terms form the structural effect, which captures differences in group-specific associations between these characteristics and changes in mental well-being, including the intercept difference. A positive WM–minority gap indicates that the minority group experienced a larger deterioration in mental well-being than the White majority.

The decomposition is estimated using COVID-19 cross-sectional weights, with standard errors adjusted for the survey design. I report decompositions using pooled coefficients as the main specification and White-majority coefficients as a robustness check.

This framework has three main limitations. First, the decomposition is descriptive rather than causal. Some explanatory variables, such as employment disruption and financial deterioration, may themselves be affected by changes in mental well-being. Second, the structural component is difficult to interpret because it may reflect heterogeneous associations, omitted variables, measurement error, or the choice of reference coefficients. Third, the Oaxaca–Blinder decomposition relies on a linear specification, which may be restrictive for variables such as housing space or the number of dependent children.

Event Study

Following the decomposition analysis, I use an event-study-style dynamic difference-in-differences specification to examine whether ethnic gaps in mental well-being changed around the onset of COVID-19. The outcome variable is the April-adjusted and inverted GHQ-12 Likert score in levels. The calendar year 2019 serves as the omitted baseline period. For the pre-pandemic UKHLS main survey, periods are defined by the calendar year of interview from 2016 to 2019, while during the pandemic periods are defined by COVID-19 survey wave.

Let M_i be an indicator equal to one if individual i belongs to the minority group, where M is defined either as BAME or as BIP depending on the comparison. Let D_t^k denote an indicator for period k. I estimate:

$$GHQ_{it} = \alpha_i + \lambda_t + \sum_{k \neq 2019} \theta_k \left(M_i \times D_t^k \right) + \varepsilon_{it}. \tag{5}$$

Here, α_i denotes individual fixed effects, which absorb time-invariant differences across individuals, including baseline ethnic differences. The period fixed effects λ_t capture shocks common to all individuals in a given period. The interaction coefficients θ_k measure the change in the minority–White majority gap in mental well-being in period k, relative to the corresponding gap in 2019:

$$\begin{aligned} \theta_k &= \Delta_k^M - \Delta_k^{WM}, \\ \Delta_k^M &= E(GHQ_{it} \mid M_i = 1, t = k) - E(GHQ_{it} \mid M_i = 1, t = 2019), \\ \Delta_k^{WM} &= E(GHQ_{it} \mid M_i = 0, t = k) - E(GHQ_{it} \mid M_i = 0, t = 2019). \end{aligned} \tag{6}$$

Since the GHQ measure is inverted so that higher values indicate better mental well-being, a negative post-pandemic coefficient indicates that the minority group experienced a relative deterioration in mental well-being compared with the White majority. For these coefficients to be interpreted as pandemic-related changes in the minority–White majority gap, the key assumption is that, in the absence of the pandemic-related shock, this gap would not have changed differentially after 2019, conditional on individual and period fixed effects. The pre-pandemic coefficients for 2016–2018 provide diagnostic evidence for this assumption. If these coefficients differ systematically from zero, this would suggest that the ethnic gap was already evolving before the pandemic, weakening the interpretation of the post-pandemic coefficients as a newly emerging pandemic-related widening of the gap.

The baseline estimates are obtained from an unweighted individual fixed-effects model because the estimation sample is a custom balanced panel linking annual UKHLS interviews to the

COVID-19 Study,[13] for which no single official survey weight is available. Standard errors are clustered at the individual level.

4 Results

Univariate Associations with Changes in Mental Well-Being

I begin by presenting descriptive evidence on changes in mental well-being and their associations with the socioeconomic and demographic characteristics discussed in Section 2. The corresponding results for BAME and BIP respondents are reported in Appendix A. These estimates are based on univariate group-specific specifications, allowing the association between each characteristic and changes in mental well-being to differ between the White majority and the minority group. They are descriptive and should not be interpreted as causal effects. Because the minority samples are smaller, the standard errors are generally larger for BAME and especially BIP respondents. As a result, although the point estimates often suggest larger declines among minority respondents, many minority–WM differences are imprecisely estimated and not statistically significant at conventional levels.

Employment and financial conditions show the clearest descriptive relationship with changes in mental well-being (A.1, A.2, A.8, and A.9). In both the BAME and BIP samples, mental well-being deteriorated most sharply among respondents who experienced job loss and among those whose financial situation worsened during the pandemic. Self-employed workers and respondents who experienced reduced working hours or furlough also exhibited relatively large declines. By contrast, the estimates for key worker status do not indicate systematically worse mental well-being among key workers relative to the reference category. This comparison should be interpreted cautiously, however, because the non-key-worker reference group also includes respondents who were not working before the pandemic. Overall, the estimated deterioration is larger for minority groups than for the White majority, with the differences more pronounced when the minority sample is restricted to BIP respondents.

[13] Respondents must be observed in 2019 and in at least one post-COVID wave.

The associations between housing conditions and changes in mental well-being are less straightforward (Appendix Tables A.3 and A.10). The results do not reveal a simple monotonic relationship between housing space and mental well-being. For the bedroom-per-person ratio, respondents in both more crowded and less crowded households often experienced larger declines than those in the intermediate category, particularly among minority respondents. A similarly uneven pattern appears for the number of other rooms. These patterns suggest that the association between housing space and mental well-being may reflect not only crowding, but also household composition, socioeconomic circumstances, and the way housing space was used during lockdown.

Household composition also appears to be related to changes in mental well-being, although the patterns differ across groups (A.6 and A.13). Larger declines are observed among single-adult households in the BAME and White-majority samples, whereas the corresponding pattern is less clear in the BIP sample. This likely reflects the limited number of single-adult households within the BIP group, which reduces the precision of the estimates. The presence of dependent children is also associated with larger declines among minority respondents, while differences across household types are generally smaller among the White majority.

The health-related variables show a mixed pattern (A.4, A.5, A.11, and A.12). Among BAME and White-majority respondents, those reporting COVID-19 symptoms experienced larger declines in mental well-being than those without such symptoms, whereas this pattern is less clear in the BIP sample. By contrast, respondents with pre-existing chronic conditions often experienced similar or smaller declines than those without the corresponding conditions. This is consistent with the view that poorer mental health among individuals with chronic conditions may primarily reflect worse baseline mental health rather than a larger deterioration after the onset of the pandemic (Rünger et al. 2025).

Finally, the demographic patterns suggest that sex and age composition may matter for ethnic differences in mental well-being changes (A.7 and A.14). There is little evidence of statistically significant BAME–WM or BIP–WM differences among women. By contrast, minority respondents, especially men, tend to exhibit larger declines in mental well-being than the White majority. Across age groups, the largest declines are concentrated among younger adults in the broader

BAME comparison, while relatively large declines are also observed among BIP respondents aged 30–69.

Taken together, these descriptive associations suggest that employment and financial conditions, household composition, health-related exposure, and demographic structure are relevant for understanding ethnic differences in changes in mental well-being. They motivate the multivariate decomposition analysis below, which examines the extent to which these factors account for the WM–minority gaps.

Decomposition and Event-Study Results

The decomposition groups the explanatory variables into six blocks: employment and financial conditions, housing conditions, health, household composition, sex, and age. These blocks follow the mechanisms discussed in Section 2 and correspond to the variables introduced in the data section. The results should be interpreted descriptively, as the decomposition separates the observed WM–minority gap into differences in characteristics and differences in group-specific associations, rather than identifying causal mechanisms.

Table 8 reports the decomposition of the WM–BAME gap in changes in mental well-being. The estimated raw gap is small and statistically insignificant, at 0.031 standard deviations. The upper panel reports the composition effect, while the lower panel reports the structural effect. In the full specification, differences in employment and financial conditions, health, and age are associated with a larger WM–BAME gap, and the total composition effect is positive and statistically significant. However, the total structural effect is small and statistically insignificant, leaving the overall raw gap small and imprecisely estimated.

The housing component is more difficult to interpret. Its composition effect is negative in the full specification, even though minority respondents are more likely than the White majority to live in crowded housing conditions. Columns (3)–(5) examine the sensitivity of this result by isolating the housing block, combining it with household composition and demographic controls, and excluding it from the decomposition. The estimated housing contribution varies substantially across specifications, suggesting that this component is not robust. Excluding the housing block does not change the main qualitative conclusion: observed characteristics are associated with a positive composition effect, while the total structural effect remains small and statistically insignificant.

Table 9 shows a different pattern when the minority group is restricted to Bangladeshi, Indian, and Pakistani respondents. The WM–BIP gap is larger and statistically significant, at 0.187 standard deviations in the full model. The total composition effect is also positive and statistically significant. Differences in employment and financial conditions and age account for an important part of the gap, while sex contributes negatively. The structural component is also positive and statistically significant, with employment and financial conditions making the largest contribution. This pattern is consistent with the possibility that similar employment and financial vulnerabilities were associated with larger deterioration in mental well-being among BIP respondents than among the White majority. In contrast to the broader BAME comparison, both differences in observed characteristics and differences in group-specific associations appear to matter for the BIP gap.

Figures 2 and 3, together with Tables B.1 and B.2, examine the dynamics of these gaps using the event-study specification. For the BAME comparison, the event-study estimates do not show a large or persistent widening of the ethnic gap after the onset of the pandemic. Although the interaction coefficient for June 2020 is negative and marginally significant, the subsequent coefficients are small and imprecisely estimated. Moreover, the pre-pandemic coefficients are not uniformly close to zero, and the joint pre-trend test is statistically significant. This suggests that the BAME–WM gap was already evolving before the pandemic, making it difficult to interpret the post-pandemic coefficients as evidence of a newly generated gap.

The event-study results for the BIP comparison are more consistent with a pandemic-related widening of the gap. The interaction coefficients become substantially negative in April 2020 and remain negative in several subsequent waves, indicating that BIP respondents experienced a relative deterioration in mental well-being compared with the White majority. Although the gap narrows somewhat later in the sample, the interaction coefficients remain below zero through the observation window. Importantly, the joint pre-trend test does not reject equality of the pre-pandemic interaction coefficients to zero, suggesting no clear evidence of differential pre-pandemic trends between BIP and White-majority respondents.

Table 8: Decomposition of the WM–BAME Gap in Changes in Mental Well-Being

	(1)	(2)	(3)	(4)	(5)
Raw WM–BAME gap	0.031	0.031	0.031	0.031	0.031
	(0.044)	(0.044)	(0.047)	(0.046)	(0.044)
Composition component attributable to:					
Employment and financial	0.017**	0.016*			0.015*
	(0.008)	(0.009)			(0.008)
Housing	-0.015*	-0.019**	0.006	-0.012	
	(0.008)	(0.009)	(0.007)	(0.009)	
Health	0.013**	0.013**			0.013**
	(0.006)	(0.007)			(0.006)
Household composition	0.007	0.007		0.008	0.005
	(0.005)	(0.006)		(0.005)	(0.005)
Sex	0.001	0.001		0.001	0.001
	(0.004)	(0.004)		(0.004)	(0.004)
Age	0.021**	0.022**		0.024**	0.017*
	(0.010)	(0.011)		(0.010)	(0.010)
Total composition effect	0.044***	0.040***	0.006	0.021**	0.052***
	(0.014)	(0.015)	(0.007)	(0.010)	(0.013)
Structural component attributable to:					
Employment and financial	0.086	0.087			0.098
	(0.061)	(0.062)			(0.061)
Housing	-0.389**	-0.385**	-0.338*	-0.390**	
	(0.176)	(0.174)	(0.174)	(0.194)	
Health	-0.019	-0.019			-0.019
	(0.036)	(0.036)			(0.036)
Household composition	-0.000	-0.000		0.001	0.014
	(0.038)	(0.040)		(0.039)	(0.035)
Sex	-0.027	-0.027		-0.028	-0.031
	(0.043)	(0.043)		(0.045)	(0.043)
Age	-0.067	-0.068		-0.063	-0.028
	(0.216)	(0.218)		(0.194)	(0.206)
Constant	0.404	0.404	0.362*	0.490	-0.054
	(0.325)	(0.325)	(0.189)	(0.321)	(0.213)
Total structural effect	-0.013	-0.009	0.025	0.010	-0.021
	(0.045)	(0.046)	(0.048)	(0.048)	(0.045)
Observations	12,533	12,533	12,533	12,533	12,533

Notes: The dependent variable is the individual change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020. Reported gaps are defined as White majority (WM) minus BAME respondents; positive values therefore indicate a larger deterioration among minority respondents relative to the White majority. Standard errors are reported in parentheses and are adjusted for the survey design, accounting for primary sampling units and strata. Columns (1) and (2) include all variable blocks. Column (1) uses pooled coefficients as the reference coefficients, while Column (2) uses White-majority coefficients. Column (3) includes only the housing block, Column (4) combines housing with household composition and demographic controls, and Column (5) excludes the housing block. The omitted categories are no change in employment status, age 70 or above, and male; the no-change employment category also includes respondents who were not working before the pandemic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Decomposition of the WM–BIP Gap in Changes in Mental Well-Being

	(1)	(2)	(3)	(4)	(5)
Raw WM–BIP gap	0.187***	0.187***	0.187***	0.187***	0.187***
	(0.060)	(0.060)	(0.063)	(0.062)	(0.061)
Composition component attributable to:					
Employment and financial	0.027**	0.024**			0.026**
	(0.012)	(0.012)			(0.012)
Housing	-0.028*	-0.030*	0.008	-0.023	
	(0.016)	(0.016)	(0.013)	(0.016)	
Health	0.015	0.015			0.016
	(0.010)	(0.010)			(0.010)
Household composition	0.015*	0.016*		0.017*	0.012
	(0.009)	(0.009)		(0.009)	(0.008)
Sex	-0.015**	-0.015**		-0.014**	-0.015**
	(0.007)	(0.007)		(0.007)	(0.007)
Age	0.039**	0.042***		0.041***	0.031**
	(0.016)	(0.016)		(0.015)	(0.014)
Total composition effect	0.054**	0.053**	0.008	0.021	0.070***
	(0.026)	(0.026)	(0.013)	(0.019)	(0.024)
Structural component attributable to:					
Employment and financial	0.186***	0.189***			0.194***
	(0.070)	(0.070)			(0.072)
Housing	0.173	0.175	0.318	0.111	
	(0.229)	(0.227)	(0.252)	(0.239)	
Health	-0.011	-0.011			-0.007
	(0.034)	(0.034)			(0.034)
Household composition	-0.095	-0.096		-0.089	-0.071
	(0.066)	(0.067)		(0.077)	(0.066)
Sex	-0.091**	-0.090**		-0.057	-0.087**
	(0.042)	(0.042)		(0.049)	(0.043)
Age	-0.267	-0.270		-0.244	-0.122
	(0.265)	(0.266)		(0.222)	(0.258)
Constant	0.237	0.237	-0.139	0.446	0.211
	(0.376)	(0.376)	(0.261)	(0.354)	(0.247)
Total structural effect	0.133**	0.134**	0.178***	0.166**	0.117*
	(0.062)	(0.062)	(0.065)	(0.065)	(0.062)
Observations	11,385	11,385	11,385	11,385	11,385

Notes: The dependent variable is the individual change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020. Reported gaps are defined as White majority (WM) minus BIP respondents; positive values therefore indicate a larger deterioration among minority respondents relative to the White majority. Standard errors are reported in parentheses and are adjusted for the survey design, accounting for primary sampling units and strata. Columns (1) and (2) include all variable blocks. Column (1) uses pooled coefficients as the reference coefficients, while Column (2) uses White-majority coefficients. Column (3) includes only the housing block, Column (4) combines housing with household composition and demographic controls, and Column (5) excludes the housing block. The omitted categories are no change in employment status, age 70 or above, and male; the no-change employment category also includes respondents who were not working before the pandemic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I further examine whether this pattern is concentrated among particular subgroups by estimating the event-study specification separately by sex and by baseline financial difficulty. Figure 4 and Figure 5 present the corresponding group-specific trajectories, while the full interaction coefficients are reported in Appendix Tables B.3 and B.4.

Figure 4 suggests that the widening of the gap is more pronounced among men. Among men, BIP respondents experience a sharp relative decline in April 2020, and the gap remains negative throughout the post-pandemic period. The corresponding pattern among women is weaker: although BIP women also experience some relative deterioration after the onset of the pandemic, the estimates are smaller, less persistent, and more imprecisely estimated. This pattern is consistent with the descriptive evidence that minority men experienced larger declines in mental well-being than the White majority.

Figure 5 shows that baseline financial difficulty is also related to the magnitude of the dynamic gap. Among respondents who reported financial difficulty in 2019, BIP respondents experience a large relative decline in the first months of the pandemic. Among respondents without baseline financial difficulty, the post-pandemic coefficients are also generally negative, but the early widening is smaller. These results suggest that pre-existing financial vulnerability amplified the relative deterioration in mental well-being among BIP respondents, although the gap is not confined to this subgroup.

Taken together, the decomposition and event-study results indicate that the broad BAME category masks substantial heterogeneity, while the BIP group experienced a larger and more persistent deterioration in mental well-being at the onset of the pandemic. The subgroup results further suggest that this deterioration was particularly pronounced among men and among respondents who were financially vulnerable before the pandemic. These subgroup estimates should be interpreted as descriptive evidence of heterogeneity rather than as definitive evidence of mechanisms, since the subgroups may differ along other observed and unobserved dimensions.

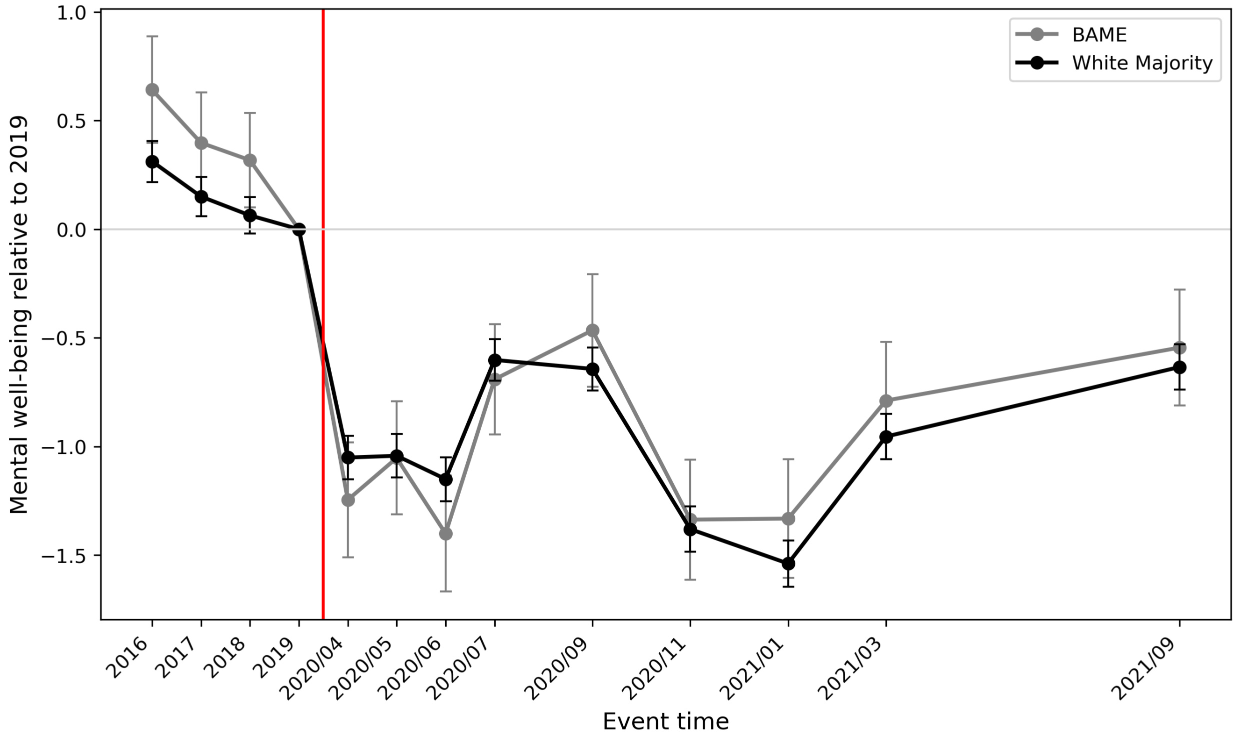

Figure 2: Event-Study Estimates of Mental Well-Being among BAME and White Majority Respondents

Notes: Data are drawn from the UKHLS main survey and the Understanding Society COVID-19 Study. The figure plots group-specific changes in the April-adjusted and inverted GHQ-12 Likert score relative to the 2019 baseline for BAME and White-majority respondents. The 2019 calendar year is the omitted baseline period and is normalized to zero for both groups. The vertical line marks the transition to the COVID-19 Study period. Estimates are obtained from an unweighted individual fixed-effects model with period fixed effects; standard errors are clustered at the individual level. Error bars show 95% confidence intervals. Higher values indicate better mental well-being.

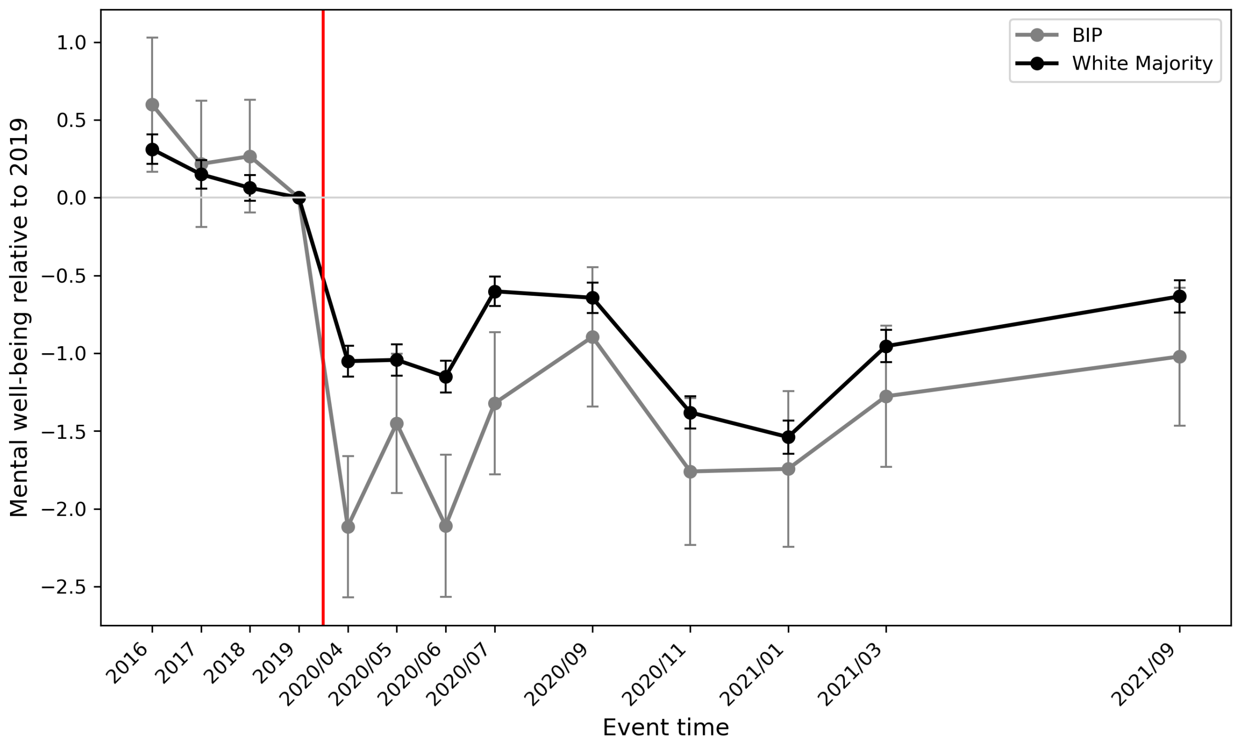

Figure 3: Event-Study Estimates of Mental Well-Being among BIP and White Majority Respondents

Notes: Data are drawn from the UKHLS main survey and the Understanding Society COVID-19 Study. The figure plots group-specific changes in the April-adjusted and inverted GHQ-12 Likert score relative to the 2019 baseline for BIP and White-majority respondents. The 2019 calendar year is the omitted baseline period and is normalized to zero for both groups. The vertical line marks the transition to the COVID-19 Study period. Estimates are obtained from an unweighted individual fixed-effects model with period fixed effects; standard errors are clustered at the individual level. Error bars show 95% confidence intervals. Higher values indicate better mental well-being.

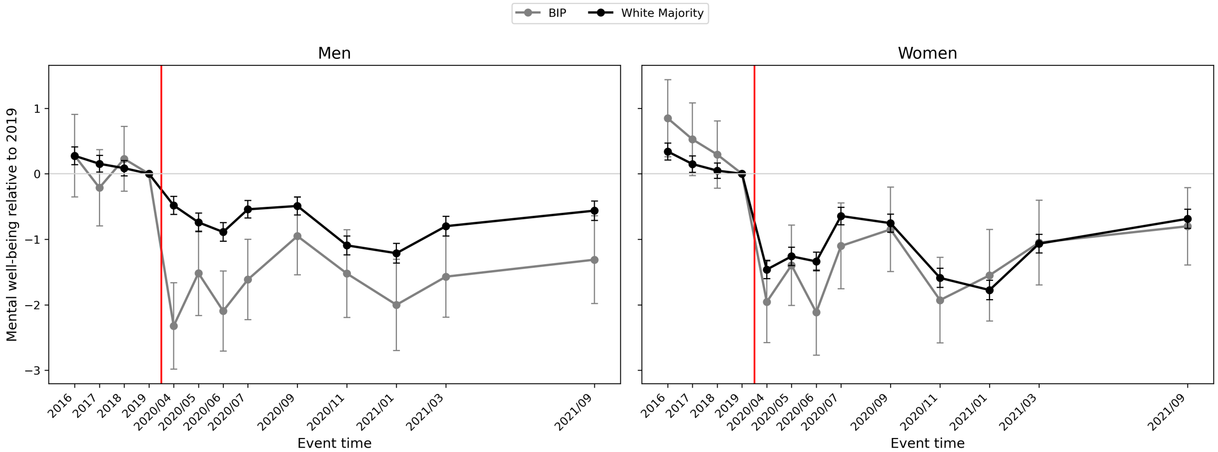

Figure 4: Event-study estimates of mental well-being among BIP and White-majority respondents by sex

Notes: The figure plots group-specific changes in the April-adjusted and inverted GHQ-12 Likert score relative to the 2019 baseline, separately for men and women. The left panel compares BIP men with White-majority men, while the right panel compares BIP women with White-majority women. The 2019 calendar year is normalized to zero for both groups. The vertical line marks the transition to the COVID-19 Study period. Estimates are obtained from unweighted individual fixed-effects models with period fixed effects. Standard errors are clustered at the individual level, and error bars indicate 95% confidence intervals. Higher values indicate better mental well-being.

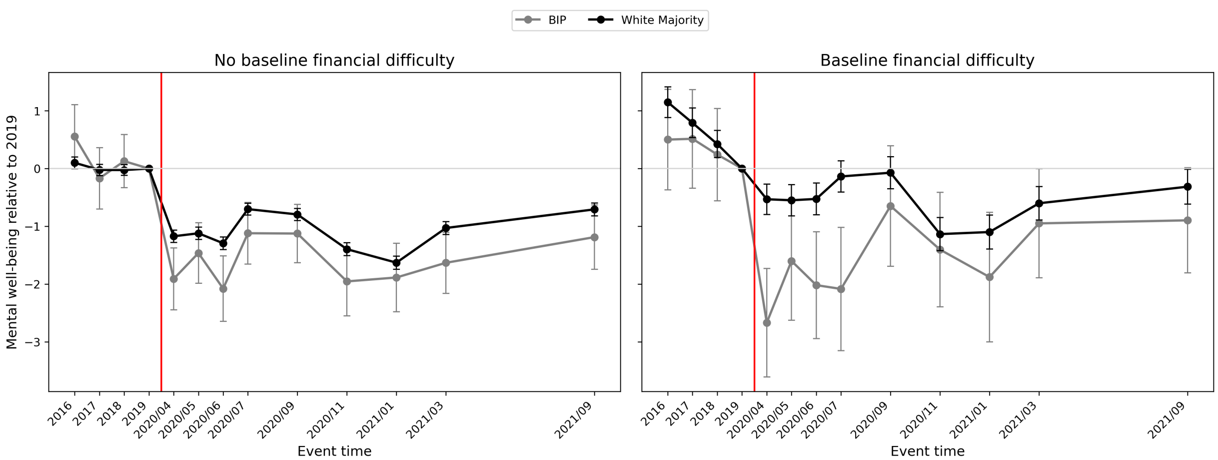

Figure 5: Event-study estimates of mental well-being among BIP and White-majority respondents by baseline financial difficulty

Notes: The figure plots group-specific changes in the April-adjusted and inverted GHQ-12 Likert score relative to the 2019 baseline, separately by baseline financial difficulty. The left panel compares BIP and White-majority respondents who did not report financial difficulty in 2019, while the right panel compares respondents who reported baseline financial difficulty. Baseline financial difficulty identifies respondents who reported that they were just about getting by, finding it quite difficult, or finding it very difficult financially in 2019. The 2019 calendar year is normalized to zero for both groups. The vertical line marks the transition to the COVID-19 Study period. Estimates are obtained from unweighted individual fixed-effects models with period fixed effects. Standard errors are clustered at the individual level, and error bars indicate 95% confidence intervals. Higher values indicate better mental well-being.

5 Discussion

This paper examined whether ethnic minority groups experienced larger declines in mental well-being at the onset of the COVID-19 pandemic in the UK. The broad BAME comparison does not provide strong evidence of a large or persistent deterioration relative to the White majority. By contrast, the BIP group experienced a larger decline in mental well-being, and the BIP–WM gap widened sharply in the early months of the pandemic. The decomposition results suggest that employment and financial conditions are important factors associated with this gap. Although the event-study coefficients become smaller in magnitude later in the sample, they remain below zero through the end of the observation window, indicating partial but incomplete convergence. The subgroup event-study results add nuance to this interpretation. The relative deterioration among BIP respondents appears more pronounced among men and among those who were financially vulnerable before the pandemic. This pattern is consistent with the view that the BIP–WM gap was related to gendered and economic dimensions of pandemic vulnerability. However, these subgroup results should still be interpreted as descriptive evidence of heterogeneity rather than as a definitive mechanism test.

Ethnic categorization is central to the interpretation of these results. The broad BAME category has the advantage of retaining smaller and otherwise underrepresented minority groups in the analysis, but it also aggregates individuals who may face very different socioeconomic circumstances, labour-market risks, household structures, and health-related vulnerabilities. This aggregation may therefore mask substantial heterogeneity in the mental well-being deterioration associated with the pandemic. The descriptive evidence illustrates this concern: respondents from "Any other mixed background" experienced the largest deterioration in mental well-being, whereas Caribbean respondents exhibited the largest improvement over the same period, although both groups are included within the broad BAME category. As a result, the BAME–White majority comparison may understate, or even obscure, the extent of ethnic inequalities in mental well-being at the onset of COVID-19.

Restricting the analysis to Bangladeshi, Indian, and Pakistani respondents reveals a more pronounced and statistically significant gap relative to the White majority. This suggests that the deterioration in mental well-being associated with the onset of the pandemic was not evenly distributed across minority ethnic groups, but was more pronounced among groups facing particular forms of social and economic vulnerability. Nevertheless, the BIP classification should also be interpreted with caution, since Bangladeshi, Indian, and Pakistani respondents are themselves heterogeneous and differ in terms of social context. The results therefore highlight the importance of moving beyond overly broad ethnic categories, while also recognizing the limitations of any aggregated classification.

A further point concerns the interpretation of changes in mental well-being rather than levels. This paper focuses on the change in mental well-being between the pre-pandemic baseline and the onset of COVID-19 in order to examine whether ethnic gaps widened after the pandemic began. This approach is useful for studying changes in ethnic inequalities, but it does not directly describe differences in the overall level of mental well-being during the pandemic. Similar changes across groups may therefore coexist with substantial differences in post-pandemic mental well-being levels. For example, the White majority and African respondents experienced relatively similar declines in mental well-being, at 0.18 and 0.20 standard deviations, respectively. However, during the COVID period, the average mental well-being level among African respondents was considerably lower than that of the White majority, at -0.49 compared with -0.15. This distinction is important because a small deterioration does not necessarily imply that a group had good mental well-being during the pandemic. Rather, it indicates only that the gap relative to the pre-pandemic baseline did not widen substantially. In this sense, the analysis should be interpreted as evidence on changes in ethnic inequalities around the onset of the pandemic, rather than as a complete assessment of ethnic differences in mental well-being levels during COVID-19.

The decomposition analysis is intended to clarify which observed characteristics are associated with the widening of the ethnic gap in changes in mental well-being. The six components included in the model—employment and financial conditions, housing conditions, health, household composition, sex, and age—are motivated by the literature reviewed in Section 2 and correspond to the main channels through which minority ethnic groups may have been differentially exposed to the pandemic shock. Interpreting the results at the component level is useful because variables within the same block are likely to be closely related. For example, employment disruption and financial deterioration are strongly connected: job loss, reduced working hours, or furlough may directly weaken household financial security, while pre-existing financial difficulties may make employment shocks more damaging for mental well-being. I therefore group them into the same component and interpret their combined contribution as capturing broader economic vulnerability. When the contribution of each component is calculated by aggregating the variable-level contributions, the corresponding standard errors account for both the variances and covariances of the underlying estimates. This allows the decomposition to capture the joint contribution of related factors, rather than treating each covariate as an independent channel.

Decomposition results

The decomposition results suggest that age composition and economic vulnerability are particularly important for the BIP–WM gap, while the broader BAME–WM gap remains small and imprecisely estimated. In the WM–BAME comparison, differences in employment and financial conditions, health, and age are associated with a positive composition component, but the overall raw gap is small because these contributions are partly offset by the structural component. When the minority group is restricted to BIP respondents, the pattern becomes more pronounced: the total composition component is positive and statistically significant, and employment and financial conditions account for an important part of the gap.

The detailed decompositions in C.2 and C.4 further suggest that financial vulnerability matters both before and during the pandemic. Pre-existing financial hardship and worsening financial circumstances after the onset of COVID-19 contribute positively to the WM–minority gap, especially in the BIP comparison. This indicates that the BIP–WM gap is associated not only with exposure to employment disruption, but also with weaker financial buffers and deterioration in financial circumstances. The structural component for employment and financial conditions is also large in the BIP comparison. This pattern is consistent with the possibility that similar employment and financial vulnerabilities were associated with larger declines in mental well-being among BIP respondents than among the White majority, although the decomposition cannot identify the underlying mechanism.

The statistically insignificant contributions of job disruption and self-employment should be interpreted carefully. These variables are closely related to financial conditions, so part of their association with mental well-being may already be captured by baseline financial hardship and financial deterioration during the pandemic. In addition, reductions in working hours or furlough may not necessarily imply worse mental health in the short run, especially under the protection of the furlough scheme (C.1 and C.3). The reference category also requires caution, since the "no change" group includes both respondents whose employment situation did not change and respondents who were not working before the pandemic.

Key worker status does not appear to be a major driver of the ethnic gap in changes in mental well-being in this analysis. Although some minority groups were disproportionately represented among key workers, respondents who remained employed as key workers did not experience significantly larger declines in mental well-being. This finding is less consistent with Qadri (2026), but closer to Serra-Sastre, Pinilla, and Kalansooriya (2026), who find no significantly worse mental-health trajectories among health and social care workers. The difference may reflect the timing of the analysis, since this paper focuses on the immediate onset of the pandemic rather than later infection peaks. Moreover, financial stability and continued social contact among employed key workers may have partly offset the negative effects of occupational exposure and workload. However, because this paper does not distinguish between different types of key workers, the estimated average effect may mask heterogeneity across sectors and occupations.

Health-related factors account for part of the WM–BAME gap in the decomposition, but their role appears limited and should be interpreted cautiously. As discussed above, part of this may reflect heterogeneity within the broad BAME category, since health exposure differs substantially across minority groups. Consistent with this, health factors do not play a comparable role in explaining the WM–BIP gap. Another concern is measurement. Lower healthcare use among some minority groups, partly due to language barriers and cultural differences, may lead to underdiagnosis of chronic conditions, causing health-related vulnerability to be underestimated (Grey et al. 2013). Moreover, chronic conditions may primarily affect baseline mental-health levels rather than the extent of deterioration after the onset of the pandemic. The detailed results show that most chronic-condition variables are not significantly associated with larger declines in mental well-being, and in some cases respondents with chronic conditions experienced smaller declines. This is consistent with Rünger et al. (2025), who suggest that poorer mental health among individuals with chronic conditions may reflect worse baseline mental health rather than a stronger pandemic-induced decline.

The housing composition component is difficult to interpret. Although BAME and BIP respondents had, on average, less bedroom space than the White majority, the housing composition component is negative in the main decomposition results. This is contrary to the expectation that more crowded housing conditions would be associated with a wider WM–minority gap during lockdown. The estimated housing composition component also varies substantially across specifications, suggesting that it is not a robust explanatory channel.

The housing structural component should also be interpreted cautiously. In the WM–BAME comparison, the structural component is negative and statistically significant, but this should not be interpreted as evidence that crowded housing was protective for minority respondents. One reason is that objective measures of housing space may not fully capture perceived crowding (Department for Levelling Up, Housing and Communities 2022). The same bedroom-per-person ratio or number of rooms may be experienced differently depending on respondents' previous housing experiences, household routines, privacy needs, and the way space was used during lockdown. The housing structural component may therefore reflect measurement limitations, heterogeneity within the broad BAME category, or instability in group-specific coefficients (C.1) rather than a clear protective effect of housing conditions.

The sex component shows different patterns across the two comparisons. In the WM–BAME comparison, both the composition and structural components are small and statistically insignificant. In the WM–BIP comparison, the sex composition component is negative and statistically significant, implying that sex composition would reduce rather than widen the BIP–WM gap. This is consistent with the descriptive evidence (A.14) showing that the BIP–WM difference is concentrated mainly among men. A plausible explanation is that BIP men faced a combination of labour-market exposure and household financial responsibilities at the onset of the pandemic. However, this mechanism is not directly observed and should therefore be interpreted cautiously.

Household composition does not appear to be a primary driver of the ethnic mental-health gap. In the WM–BAME comparison, both the composition and structural components are small and statistically insignificant. In the WM–BIP comparison, household composition contributes positively in some specifications, but the result is not robust. This suggests that household structure may have shaped exposure to pandemic-related pressures, but it does not provide a stable explanation for the BIP–WM gap.

Event Study

The event-study results for the broad BAME comparison do not provide strong evidence of a large or persistent widening of the mental-health gap after the onset of the pandemic. Although the descriptive trend in Figure 1 suggests a visible decline in mental well-being among BAME respondents, the event-study estimates indicate that the BAME–WM gap changed only modestly relative to the 2019 baseline. Moreover, the significant joint pre-trend test suggests that the BAME–WM gap was already evolving before the pandemic. The post-pandemic coefficients therefore do not provide clean evidence of a newly emerging and persistent widening of the BAME–WM gap after the onset of COVID-19.

The difference between Figure 1 and the event-study estimates likely reflects differences in empirical design. Figure 1 presents weighted descriptive group means indexed to 2016 and captures the overall evolution of mental well-being across groups. In contrast, the event study uses individual fixed effects, normalizes 2019 as the omitted baseline, and estimates whether the ethnic gap changed relative to the immediate pre-pandemic period after accounting for common time shocks. In addition, the event-study estimates are based on an unweighted custom balanced panel, whereas Figure 1 uses survey weights. These differences mean that Figure 1 is useful for describing broad trends, while the event study provides a more demanding test of whether the BAME–WM gap widened after the onset of COVID-19. The broad BAME classification may also attenuate the estimated gap by pooling groups with different pre-pandemic conditions and pandemic experiences.

By contrast, the BIP event-study results provide clearer evidence of a widening ethnic mental-health gap after the onset of the pandemic. Unlike the broad BAME comparison, the joint pre-trend test for the BIP group is not statistically significant, suggesting no clear evidence of differential pre-pandemic trends between BIP and WM respondents. After the onset of COVID-19, however, the BIP–WM interaction coefficients become negative and statistically significant in the early months of the pandemic, indicating that BIP respondents experienced a larger relative decline in mental well-being than the White majority. This pattern is consistent with the decomposition results, which show that employment and financial conditions are important factors associated with the BIP–WM gap.

The later narrowing of the gap may also be consistent with an economic-vulnerability interpretation. The reopening of parts of the service sector in July 2020, followed by the Eat Out to Help Out scheme in August (Government 2021), may have temporarily reduced employment and income uncertainty in sectors where some BIP respondents, particularly Pakistani and Bangladeshi men, were more exposed. This could help explain why the BIP–WM interaction coefficients become smaller and less precisely estimated later in the summer. However, this interpretation should be treated cautiously, since the event-study design does not directly identify the effect of reopening policies or sector-specific labour-market changes.

By November 2020, the second national lockdown represented a renewed negative shock to mental well-being more generally. However, the event-study estimates do not suggest a statistically significant additional widening of the BIP–WM gap in that wave. This may indicate that the second lockdown worsened mental well-being across the population, rather than producing a further disproportionate deterioration among BIP respondents. Overall, the BIP event-study results suggest a sharp initial widening of the mental-health gap after the onset of COVID-19, followed by partial but incomplete convergence.

Limitations

Several limitations should be noted when interpreting the findings. First, the results should not be interpreted as causal estimates of the effect of COVID-19 on mental well-being. Although the analysis examines changes in mental well-being between the pre-pandemic baseline and the onset of the pandemic, the empirical design does not observe the counterfactual trajectory of mental well-being in a world without COVID-19. The analysis therefore cannot directly compare mental health outcomes with and without the pandemic. In addition, some explanatory factors included in the decomposition may be endogenous to changes in mental well-being. Employment disruption, financial deterioration, and household stress may contribute to poorer mental health, but worsening mental health may also affect labour-market attachment, perceived financial strain, and survey responses. The decomposition results should therefore be interpreted as descriptive evidence on factors associated with WM–minority gaps in changes in mental well-being, rather than as causal mechanisms.

Second, the analysis is limited by sample size and ethnic categorization. Although the BIP classification is more informative than the broad BAME category in this analysis, it still aggregates Bangladeshi, Indian, and Pakistani respondents, who may differ substantially in socioeconomic circumstances, occupational exposure, household structure, and pandemic experiences. In addition, the sample sizes for some minority groups are relatively small, which increases statistical uncertainty and limits the extent to which the analysis can examine more disaggregated ethnic categories.

Third, the analysis is subject to measurement limitations. Mental well-being is measured using self-reported GHQ-12 responses, and some explanatory variables capture reported circumstances rather than objective conditions. For example, chronic conditions are based on reported diagnoses and may be affected by differences in healthcare access, diagnosis, and survey reporting. Similarly, the housing variables capture objective space, such as bedrooms per person and the number of other rooms, but they do not fully measure perceived crowding, privacy, housing quality, or access to outdoor space.

Fourth, the Oaxaca–Blinder decomposition relies on a linear specification. If the relationship between mental well-being and the explanatory factors is non-linear, the estimated contribution of different component blocks may be inaccurate. This concern is particularly relevant for variables such as housing space and the number of dependent children, where the descriptive results do not suggest a simple monotonic relationship with changes in mental well-being. Although the model includes quadratic terms for housing space and dependent children, these terms may still not fully capture the true functional form of the relationship.

Fifth, the decomposition analysis focuses only on the initial onset of the pandemic, using the change in mental well-being between the 2019 baseline and April 2020. This choice is useful because it captures the immediate shock of COVID-19 and the first national lockdown, while limiting the influence of later policy changes, behavioral adaptation, and other post-onset disturbances. However, it also means that the decomposition results should not be interpreted as explaining the persistence or later evolution of ethnic mental-health gaps. Those dynamics are instead examined through the event-study analysis.

Sixth, the structural component of the Oaxaca–Blinder decomposition is difficult to interpret in detail. Unlike the composition component, which captures differences in observed characteristics across groups, the structural component reflects differences in group-specific coefficients. These differences may capture heterogeneous associations between observed characteristics and changes in mental well-being, but they may also reflect unobserved characteristics, model specification, omitted variables, measurement error, or the choice of reference coefficients. These components should therefore not be interpreted as clear causal mechanisms.

Finally, the event-study estimates also have limitations. Because the event-study sample is a custom balanced panel linking annual UKHLS interviews to the COVID-19 Study, no single official survey weight is available for this specific panel structure. As a result, the event-study estimates should be interpreted as within-sample dynamic patterns rather than fully population-representative estimates. This is potentially important because the UKHLS and COVID-19 surveys rely on a clustered and stratified sampling design with unequal selection probabilities and differential non-response. If sampling or attrition patterns are correlated with ethnicity and mental well-being, the unweighted event-study estimates may understate or overstate the true population-level changes in the ethnic mental-health gap.

Taken together, these limitations suggest that the findings should be interpreted with caution. Nevertheless, the main result remains informative: while the broader BAME–WM comparison does not show a large or robust widening of the mental-health gap, the BIP–WM comparison points to a more pronounced deterioration among Bangladeshi, Indian, and Pakistani respondents after the onset of COVID-19.

6 Conclusion

This paper examined whether ethnic minority groups in the UK experienced larger declines in mental well-being at the onset of the COVID-19 pandemic. Using data from Understanding Society and the Understanding Society COVID-19 Study, I compared changes in GHQ-12 mental well-being between the 2019 baseline and April 2020, and used the event-study analysis to trace the subsequent evolution of ethnic gaps during the pandemic. The analysis combined Oaxaca–Blinder decompositions with an event-study approach in order to examine both the observed socioeconomic and demographic factors associated with ethnic gaps and the dynamic evolution of these gaps after the onset of COVID-19.

The results show that the broad BAME–White majority comparison does not provide strong evidence of a large or persistent widening of the mental-health gap. However, this aggregate comparison masks substantial heterogeneity across minority ethnic groups. When the analysis focuses on Bangladeshi, Indian, and Pakistani respondents, the results point to a clearer and more pronounced deterioration in mental well-being relative to the White majority. The decomposition results suggest that age composition and economic vulnerability, particularly employment and financial conditions, are important factors associated with the WM–BIP gap. The event-study estimates further show a sharp widening of the BIP–WM gap in the first months of the pandemic, followed by partial but incomplete convergence. Subgroup event-study results suggest that this relative deterioration was particularly pronounced among men and among respondents who were financially vulnerable before the pandemic.

These findings highlight the importance of ethnic categorization in the study of health inequalities. Broad minority classifications can be useful for retaining smaller groups in the analysis, but they may also obscure important differences in social context, economic exposure, and pandemic vulnerability. The results therefore suggest that the deterioration in mental well-being associated with the onset of COVID-19 was not evenly distributed across all minority ethnic groups, but was more pronounced among groups facing specific forms of socioeconomic vulnerability.

Several limitations should be kept in mind. The results should not be interpreted as causal estimates of the effect of COVID-19 on mental well-being, since the analysis does not observe the counterfactual trajectory of mental health in the absence of the pandemic. In addition, the Oaxaca–Blinder decomposition relies on a linear specification, and the structural component is difficult to interpret as a clear mechanism. The analysis is also limited by small minority-group sample sizes, measurement constraints, and the need to aggregate ethnic groups. Finally, the event-study estimates are based on an unweighted custom balanced panel and should therefore be interpreted as within-sample dynamic patterns rather than fully population-representative estimates.

Overall, the paper provides evidence that the onset of the COVID-19 pandemic was associated with a widening mental-health gap between BIP respondents and the White majority in the UK. The findings suggest that policies aimed at reducing ethnic inequalities in mental health should pay attention not only to healthcare access, but also to the design, timing, and coverage of economic support. Even with furlough and self-employment support schemes, economically vulnerable groups may remain exposed to financial insecurity and associated mental-health risks. Future research should examine whether these gaps persisted beyond the early phase of the pandemic and investigate the mechanisms linking economic vulnerability to mental well-being deterioration among Bangladeshi, Indian, and Pakistani respondents.

A Univariate Associations with Changes in Mental Well-Being

Table A.1: Changes in Mental Well-Being by BAME Status and Employment Characteristics

	BAME	BAME	BAME	WM	WM	WM	p-value
Change in employment status							
No change	-0.17***			-0.16***			[0.92]
	(0.06)			(0.02)			
Reduction	-0.35***			-0.20***			[0.16]
	(0.10)			(0.03)			
Job loss	-0.79***			-0.38***			[0.17]
	(0.29)			(0.08)			
Key worker, no employment disruption							
Yes		-0.21*			-0.15***		[0.61]
		(0.11)			(0.04)		
No		-0.26***			-0.19***		[0.25]
		(0.06)			(0.02)		
Self-employed only at baseline							
Yes			-0.50**			-0.25***	[0.28]
			(0.23)			(0.05)	
No			-0.22***			-0.17***	[0.37]
			(0.05)			(0.02)	
Observations	2211	2209	2212	12575	12572	12581	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. "No change" includes both respondents who were not working before the pandemic and respondents whose employment situation did not change between the 2019 baseline and the April 2020 COVID-19 Study. "Reduction" includes respondents who were furloughed or experienced reduced working hours, while "Job loss" indicates respondents who were employed before the pandemic but were no longer employed in April 2020. "Key worker" is defined only for respondents who were employed before the pandemic and experienced no employment change; all other respondents are coded as non-key workers. "Self-employed" refers to respondents who reported being self-employed only before the pandemic. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Changes in Mental Well-Being by BAME Status and Financial Conditions

	BAME	BAME	WM	WM	p-value
Financial difficulties before COVID					
No	-0.23***		-0.18***		[0.42]
	(0.06)		(0.02)		
Yes	-0.28***		-0.17***		[0.32]
	(0.11)		(0.04)		
Change in financial status					
Better		-0.18		0.00	[0.12]
		(0.11)		(0.04)	
No change		-0.15**		-0.19***	[0.56]
		(0.07)		(0.02)	
Worse		-0.63***		-0.41***	[0.15]
		(0.15)		(0.05)	
Observations	2200	2198	12570	12558	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Baseline financial difficulties are defined using respondents' self-assessed financial circumstances in 2019. Respondents who reported that they were "just about getting by," "finding it quite difficult," or "finding it very difficult" are classified as experiencing baseline financial difficulties. Worsened financial situation indicates that respondents reported being worse off financially in April 2020 than before the pandemic. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Changes in Mental Well-Being by BAME Status and Housing Space

	BAME	BAME	WM	WM	p-value
Bedroom-per-person ratio					
<1 per person	-0.30***		-0.20***		[0.16]
	(0.06)		(0.04)		
1-<2 per person	-0.21**		-0.15***		[0.54]
	(0.09)		(0.02)		
≥2 per person	-0.31		-0.22***		[0.66]
	(0.20)		(0.04)		
Other rooms at baseline					
0-1 rooms		-0.23***		-0.17***	[0.53]
		(0.08)		(0.04)	
2-3 rooms		-0.26***		-0.19***	[0.37]
		(0.08)		(0.02)	
4+ rooms		-0.49*		-0.19***	[0.29]
		(0.28)		(0.04)	
Observations	2126	2126	12387	12387	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. The bedroom-per-person ratio is measured at the 2019 baseline and captures sleeping-space constraints. The number of other rooms refers to rooms in the accommodation excluding kitchens and bathrooms, and captures broader living space available within the household. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Changes in Mental Well-Being by BAME Status: COVID-19 Symptoms and Chronic Conditions I

	BAME	BAME	BAME	WM	WM	WM	p-value
COVID symptoms							
No	-0.24***			-0.16***			[0.23]
	(0.06)			(0.02)			
Yes	-0.33**			-0.31***			[0.90]
	(0.13)			(0.07)			
Respiratory chronic condition							
No		-0.25***			-0.18***		[0.26]
		(0.06)			(0.02)		
Yes		-0.25*			-0.15***		[0.50]
		(0.15)			(0.04)		
Cardiovascular chronic condition							
No			-0.27***			-0.19***	[0.21]
			(0.06)			(0.02)	
Yes			-0.11			-0.13***	[0.91]
			(0.12)			(0.05)	
Observations	2210	2207	2207	12578	12537	12537	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Health-condition indicators follow the definitions described in Section 2 and Table 5. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Changes in Mental Well-Being by BAME Status: Chronic Conditions II

	BAME	BAME	BAME	WM	WM	WM	p-value
Endocrine chronic condition							
No	-0.29***			-0.18***			[0.08]
	(0.06)			(0.02)			
Yes	0.02			-0.14***			[0.29]
	(0.15)			(0.04)			
Arthritis							
No		-0.28***			-0.19***		[0.14]
		(0.05)			(0.02)		
Yes		-0.01			-0.08*		[0.79]
		(0.25)			(0.04)		
Other chronic condition							
No			-0.29***			-0.18***	[0.05]
			(0.06)			(0.02)	
Yes			-0.08			-0.17***	[0.59]
			(0.17)			(0.05)	
Observations	2207	2207	2207	12537	12537	12537	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Health-condition indicators follow the definitions described in Section 2 and Table 5. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Changes in Mental Well-Being by BAME Status and Household Composition

	BAME	BAME	WM	WM	p-value
Household occupancy at baseline					
Multi-adult	-0.20***		-0.17***		[0.56]
	(0.05)		(0.02)		
Single-adult	-0.49***		-0.21***		[0.15]
	(0.19)		(0.05)		
Dependent children at baseline					
No kids		-0.17***		-0.16***	[0.81]
		(0.07)		(0.02)	
1-2 dep kids		-0.24***		-0.22***	[0.80]
		(0.07)		(0.03)	
$\geq$3 dep kids		-0.32***		-0.19*	[0.39]
		(0.10)		(0.10)	
Observations	2202	1954	12575	10910	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Household composition is measured at the 2019 baseline. "Single-adult household" identifies respondents living in households with only one adult. The dependent-children categories report the number of dependent children in the household at baseline. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Changes in Mental Well-Being by BAME Status and Demographic Characteristics

	BAME	BAME	WM	WM	p-value
Sex					
Male	-0.25***		-0.08***		[0.05]
	(0.08)		(0.02)		
Female	-0.26***		-0.26***		[0.91]
	(0.07)		(0.02)		
Age group					
16-29		-0.41***		-0.27***	[0.28]
		(0.12)		(0.05)	
30-49		-0.25***		-0.20***	[0.60]
		(0.08)		(0.03)	
50-69		-0.12		-0.13***	[0.92]
		(0.10)		(0.03)	
70+		-0.18		-0.15***	[0.83]
		(0.11)		(0.03)	
Observations	2201	2202	12575	12575	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BAME and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Sex and age are measured at the 2019 baseline. "Female" indicates respondents identified as women. Age groups are defined using respondents' age at baseline, with respondents aged 70 or above serving as the oldest age category. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BAME and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Changes in Mental Well-Being by BIP Status and Employment Characteristics

	BIP	BIP	BIP	WM	WM	WM	p-value
Change in employment status							
No change	-0.33***			-0.16***			[0.02]
	(0.08)			(0.02)			
Reduction	-0.30**			-0.20***			[0.44]
	(0.14)			(0.03)			
Job loss	-0.87***			-0.38***			[0.03]
	(0.23)			(0.08)			
Key worker, no employment disruption							
Yes		-0.24**			-0.15***		[0.35]
		(0.10)			(0.04)		
No		-0.38***			-0.19***		[0.01]
		(0.08)			(0.02)		
Self-employed only at baseline							
Yes			-0.47***			-0.25***	[0.19]
			(0.18)			(0.05)	
No			-0.34***			-0.17***	[0.01]
			(0.07)			(0.02)	
Observations	800	799	801	12575	12572	12581	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. "No change" includes both respondents who were not working before the pandemic and respondents whose employment situation did not change between the 2019 baseline and the April 2020 COVID-19 Study. "Reduction" includes respondents who were furloughed or experienced reduced working hours, while "Job loss" indicates respondents who were employed before the pandemic but were no longer employed in April 2020. "Key worker" is defined only for respondents who were employed before the pandemic and experienced no employment change; all other respondents are coded as non-key workers. "Self-employed" refers to respondents who reported being self-employed only before the pandemic. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: Changes in Mental Well-Being by BIP Status and Financial Conditions

	BIP	BIP	WM	WM	p-value
Financial difficulties at baseline					
No	-0.27***		-0.18***		[0.17]
	(0.07)		(0.02)		
Yes	-0.51***		-0.17***		[0.00]
	(0.11)		(0.04)		
Change in financial status					
Better		-0.24		0.00	[0.12]
		(0.17)		(0.04)	
No change		-0.20***		-0.19***	[0.80]
		(0.07)		(0.02)	
Worse		-0.82***		-0.41***	[0.01]
		(0.16)		(0.05)	
Observations	797	795	12570	12558	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Baseline financial difficulties are defined using respondents' self-assessed financial circumstances in 2019. Respondents who reported that they were "just about getting by," "finding it quite difficult," or "finding it very difficult" are classified as experiencing baseline financial difficulties. Worsened financial situation indicates that respondents reported being worse off financially in April 2020 than before the pandemic. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Changes in Mental Well-Being by BIP Status and Housing Space

	BIP	BIP	WM	WM	p-value
Bedroom-per-person ratio at baseline					
<1 per person	-0.39***		-0.20***		[0.04]
	(0.09)		(0.04)		
1-<2 per person	-0.28**		-0.15***		[0.27]
	(0.13)		(0.02)		
≥2 per person	-0.40**		-0.22***		[0.23]
	(0.16)		(0.04)		
Other rooms at baseline					
0-1 rooms		-0.21**		-0.17***	[0.68]
		(0.09)		(0.04)	
2-3 rooms		-0.49***		-0.19***	[0.00]
		(0.10)		(0.02)	
4+ rooms		-0.39		-0.19***	[0.38]
		(0.25)		(0.04)	
Observations	759	760	12387	12387	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. The bedroom-per-person ratio is measured at the 2019 baseline and captures sleeping-space constraints. The number of other rooms refers to rooms in the accommodation excluding kitchens and bathrooms, and captures broader living space available within the household. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.11: Changes in Mental Well-Being by BIP Status: COVID-19 Symptoms and Chronic Conditions I

	BIP	BIP	BIP	WM	WM	WM	p-value
COVID symptoms							
No	-0.35***			-0.16***			[0.01]
	(0.08)			(0.02)			
Yes	-0.37*			-0.31***			[0.76]
	(0.20)			(0.07)			
Respiratory chronic condition							
No		-0.34***			-0.18***		[0.02]
		(0.07)			(0.02)		
Yes		-0.57*			-0.15***		[0.13]
		(0.31)			(0.04)		
Cardiovascular chronic condition							
No			-0.35***			-0.19***	[0.01]
			(0.07)			(0.02)	
Yes			-0.42**			-0.13***	[0.11]
			(0.20)			(0.05)	
Observations	800	799	799	12578	12537	12537	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Health-condition indicators follow the definitions described in Section 2 and Table 5. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.12: Changes in Mental Well-Being by BIP Status: Chronic Conditions II

	BIP	BIP	BIP	WM	WM	WM	p-value
Endocrine chronic condition							
No	-0.38***			-0.18***			[0.01]
	(0.08)			(0.02)			
Yes	-0.24			-0.14***			[0.56]
	(0.18)			(0.04)			
Arthritis							
No		-0.34***			-0.19***		[0.02]
		(0.07)			(0.02)		
Yes		-0.61			-0.08*		[0.11]
		(0.38)			(0.04)		
Other chronic condition							
No			-0.38***			-0.18***	[0.00]
			(0.07)			(0.02)	
Yes			-0.14			-0.17***	[0.92]
			(0.30)			(0.05)	
Observations	799	799	799	12537	12537	12537	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Health-condition indicators follow the definitions described in Section 2 and Table 5. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.13: Changes in Mental Well-Being by BIP Status and Household Composition

	BIP	BIP	WM	WM	p-value
Household occupancy at baseline					
Multi-occupancy	-0.36***		-0.17***		[0.00]
	(0.07)		(0.02)		
Single	-0.17		-0.21***		[0.85]
	(0.25)		(0.05)		
Dependent children at baseline					
No kids		-0.33***		-0.16***	[0.05]
		(0.10)		(0.02)	
1-2 dep kids		-0.34***		-0.22***	[0.20]
		(0.10)		(0.03)	
$\geq$3 dep kids		-0.47***		-0.19*	[0.11]
		(0.16)		(0.10)	
Observations	798	764	12575	10910	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Household composition is measured at the 2019 baseline. “Single-adult household” identifies respondents living in households with only one adult. The dependent-children categories report the number of dependent children in the household at baseline. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.14: Changes in Mental Well-Being by BIP Status and Demographic Characteristics

	BIP	BIP	WM	WM	p-value
Sex					
Male	-0.34***		-0.08***		[0.00]
	(0.10)		(0.02)		
Female	-0.36***		-0.26***		[0.26]
	(0.09)		(0.02)		
Age group					
16-29		-0.26**		-0.27***	[0.90]
		(0.10)		(0.05)	
30-49		-0.38***		-0.20***	[0.08]
		(0.11)		(0.03)	
50-69		-0.49***		-0.13***	[0.00]
		(0.12)		(0.03)	
70+		-0.40**		-0.15***	[0.14]
		(0.19)		(0.03)	
Observations	798	798	12575	12575	

Notes: Data are drawn from the 2019 UKHLS main survey linked to the April 2020 Understanding Society COVID-19 Study. The table reports survey-weighted predictive margins from separate univariate regressions estimated for BIP and White-majority respondents. The outcome variable is the individual change in the standardized, April-adjusted, and inverted GHQ-12 Likert score between 2019 and April 2020. Higher values indicate better mental well-being; negative changes indicate deterioration. Sex and age are measured at the 2019 baseline. "Female" indicates respondents identified as women. Age groups are defined using respondents' age at baseline, with respondents aged 70 or above serving as the oldest age category. Standard errors, reported in parentheses, account for stratification and clustering at the PSU level. The final column reports p-values from tests of equality between the BIP and White-majority predictive margins within each category. Stars indicate whether the within-group predictive margin differs from zero. Observations are unweighted cell counts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Event study estimates of the minority and White majority gap in mental well-being

Table B.1: Event-study estimates of the BAME–White majority gap in mental well-being

Period	Event-study coefficient	p-value
2019 (base)	0.00	
2016	0.33**	0.01
2017	0.25*	0.05
2018	0.25**	0.03
2020/04	-0.19	0.18
2020/05	-0.01	0.94
2020/06	-0.25*	0.08
2020/07	-0.09	0.52
2020/09	0.18	0.21
2020/11	0.04	0.77
2021/01	0.21	0.17
2021/03	0.17	0.26
2021/09	0.09	0.54
Joint pre-trend		0.05

Notes: The table reports event-study interaction coefficients from an unweighted individual fixed-effects model. Each coefficient measures the change in the BAME–White majority mental well-being gap in the corresponding period relative to the 2019 baseline. The outcome is the April-adjusted and inverted GHQ-12 Likert score, so negative coefficients indicate that BAME respondents experienced a relative deterioration in mental well-being compared with the White majority. Standard errors are clustered at the individual level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The row "Joint pre-trend" reports the p-value from the joint test of whether all pre-pandemic interaction coefficients are equal to zero.

Table B.2: Event-study estimates of the BIP–White majority gap in mental well-being

Period	Event-study coefficient	p-value
2019 (base)	0.00	
2016	0.29	0.20
2017	0.07	0.75
2018	0.20	0.28
2020/04	-1.06***	0.00
2020/05	-0.41*	0.08
2020/06	-0.96***	0.00
2020/07	-0.72***	0.00
2020/09	-0.25	0.28
2020/11	-0.38	0.12
2021/01	-0.20	0.43
2021/03	-0.32	0.18
2021/09	-0.39*	0.10
Joint pre-trend		0.48

Notes: The table reports event-study interaction coefficients from an unweighted individual fixed-effects model. Each coefficient measures the change in the BIP–White majority mental well-being gap in the corresponding period relative to the 2019 baseline. The outcome is the April-adjusted and inverted GHQ-12 Likert score, so negative coefficients indicate that BIP respondents experienced a relative deterioration in mental well-being compared with the White majority. Standard errors are clustered at the individual level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The row "Joint pre-trend" reports the p-value from the joint test of whether all pre-pandemic interaction coefficients are equal to zero.

Table B.3: Event-study estimates of the BIP–White majority gap in mental well-being by sex

Period	Male coefficient	Male p-value	Female coefficient	Female p-value
2019 (base)	0.00		0.00	
2016	0.00	1.00	0.51*	0.10
2017	-0.37	0.23	0.38	0.19
2018	0.14	0.58	0.24	0.36
2020/04	-1.84***	0.00	-0.49	0.13
2020/05	-0.78**	0.02	-0.14	0.67
2020/06	-1.21***	0.00	-0.78**	0.02
2020/07	-1.07***	0.00	-0.46	0.18
2020/09	-0.46	0.14	-0.10	0.78
2020/11	-0.43	0.22	-0.34	0.32
2021/01	-0.79**	0.03	0.23	0.54
2021/03	-0.77**	0.02	0.02	0.96
2021/09	-0.75**	0.03	-0.11	0.71
Joint pre-trend		0.38		0.39

Notes: The table reports event-study interaction coefficients for the BIP–White majority gap in mental well-being, estimated separately by sex. The male coefficients compare BIP men with White-majority men, while the female coefficients compare BIP women with White-majority women. The 2019 calendar year is omitted as the baseline period. The outcome is the April-adjusted and inverted GHQ-12 Likert score, so higher values indicate better mental well-being. A negative coefficient indicates that BIP respondents experienced a relative deterioration in mental well-being compared with the White majority within the same sex group. Standard errors are clustered at the individual level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The row "Joint pre-trend" reports the p-value from the joint test of whether all pre-pandemic interaction coefficients are equal to zero.

Table B.4: Event-study estimates of the BIP–White majority gap in mental well-being by baseline financial difficulty

Period	No difficulty coefficient	No difficulty p-value	Difficulty coefficient	Difficulty p-value
2019 (base)	0.00		0.00	
2016	0.45	0.11	-0.65	0.16
2017	-0.15	0.60	-0.28	0.54
2018	0.15	0.52	-0.18	0.67
2020/04	-0.74***	0.01	-2.14***	0.00
2020/05	-0.34	0.21	-1.05*	0.05
2020/06	-0.79***	0.01	-1.49***	0.00
2020/07	-0.42	0.13	-1.95***	0.00
2020/09	-0.33	0.21	-0.58	0.30
2020/11	-0.56*	0.07	-0.27	0.61
2021/01	-0.26	0.41	-0.78	0.19
2021/03	-0.60**	0.03	-0.35	0.49
2021/09	-0.48*	0.10	-0.58	0.23
Joint pre-trend		0.18		0.58

Notes: The table reports event-study interaction coefficients for the BIP–White majority gap in mental well-being, estimated separately by baseline financial difficulty. The "No difficulty" columns compare BIP respondents with White-majority respondents among individuals who did not report financial difficulty in 2019, while the "Difficulty" columns compare BIP respondents with White-majority respondents among individuals who reported baseline financial difficulty in 2019. The 2019 calendar year is omitted as the baseline period. The outcome is the April-adjusted and inverted GHQ-12 Likert score, so higher values indicate better mental well-being. A negative coefficient indicates that BIP respondents experienced a relative deterioration in mental well-being compared with the White majority within the same baseline financial-difficulty group. Standard errors are clustered at the individual level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The row "Joint pre-trend" reports the p-value from the joint test of whether all pre-pandemic interaction coefficients are equal to zero.

C Multivariate Regressions and Detailed Oaxaca–Blinder Decompositions

Table C.1: Multivariate Regression Estimates for Changes in Mental Well-Being: BAME and White Majority

	(1) Pooled model	(2) BAME	(3) WM
BAME	0.013		
	(0.053)		
Employment and financial factors			
Reduced hours or furlough	0.018	-0.043	0.026
	(0.040)	(0.096)	(0.042)
Job loss	-0.197**	-0.322	-0.163*
	(0.098)	(0.238)	(0.094)
Key worker, no employment disruption	0.039	0.012	0.040
	(0.049)	(0.107)	(0.052)
Self-employed only at baseline	-0.053	0.011	-0.063
	(0.053)	(0.109)	(0.056)
Baseline financial difficulties	-0.083*	-0.064	-0.088*
	(0.046)	(0.110)	(0.047)
Financial situation improved	0.216***	0.060	0.237***
	(0.046)	(0.118)	(0.048)
Financial situation worsened	-0.219***	-0.329***	-0.207***
	(0.043)	(0.116)	(0.044)
Housing			

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	(1) Pooled model	(2) BAME	(3) WM
Bedroom-per-person ratio	-0.075	0.179	-0.114
	(0.091)	(0.204)	(0.096)
Bedroom-per-person ratio squared	0.009	-0.047	0.018
	(0.023)	(0.050)	(0.024)
Number of other rooms	-0.032	0.125	-0.038
	(0.030)	(0.144)	(0.030)
Number of other rooms squared	0.004	-0.025	0.004
	(0.003)	(0.024)	(0.003)
Health			
Reported COVID-19 symptoms	-0.090*	-0.122	-0.086*
	(0.046)	(0.119)	(0.048)
Respiratory condition	-0.004	-0.027	0.002
	(0.050)	(0.145)	(0.051)
Cardiovascular condition	0.009	-0.102	0.019
	(0.044)	(0.135)	(0.044)
Endocrine condition	0.007	0.195	-0.018
	(0.050)	(0.196)	(0.048)
Arthritis	0.123**	0.221	0.115**
	(0.056)	(0.295)	(0.051)

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	(1) Pooled model	(2) BAME	(3) WM
Other chronic condition	0.032	0.086	0.022
	(0.049)	(0.151)	(0.049)
Household composition			
Single-adult household	0.064	0.314*	0.041
	(0.132)	(0.181)	(0.139)
Number of dependent children	-0.000	-0.023	0.002
	(0.048)	(0.090)	(0.051)
Number of dependent children squared	-0.015	-0.007	-0.016
	(0.014)	(0.023)	(0.015)
Single-adult household × children	-0.185	-0.381	-0.187
	(0.242)	(0.241)	(0.311)
Single-adult household × children squared	0.099	0.121**	0.107
	(0.088)	(0.058)	(0.126)
Demographics			
Female	-0.187***	-0.143*	-0.195***
	(0.031)	(0.079)	(0.032)
Age 16–29	-0.123*	-0.088	-0.127*
	(0.074)	(0.180)	(0.077)
Age 30–49	-0.037	0.053	-0.050

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	(1) Pooled model	(2) BAME	(3) WM
	(0.074)	(0.173)	(0.077)
Age 50–69	0.048	0.090	0.039
	(0.051)	(0.167)	(0.051)
Constant	0.065	-0.292	0.111
	(0.105)	(0.228)	(0.110)
Observations	12,533	1,866	10,667

Notes: The dependent variable is the individual change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020. Higher values indicate better mental well-being; negative coefficients indicate larger deterioration in mental well-being. Column (1) reports estimates from the multivariate pooled specification. Columns (2) and (3) report estimates from group-specific specifications for the minority group and the White majority, respectively. In Column (1), the BAME coefficient is defined relative to the White majority. All models use COVID-19 cross-sectional survey weights. Standard errors, reported in parentheses, account for the survey design. The omitted categories are no change in employment status, age 70 or above, and male; the no-change employment category also includes respondents who were not working before the pandemic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2: Detailed Oaxaca–Blinder Decomposition of the WM–BAME Gap in Changes in Mental Well-Being

	(1) Pooled ref.	(2) WM ref.
Raw WM–BAME gap	0.031	0.031
	(0.044)	(0.044)
Composition effect attributable to:		
Employment factors		
Reduced hours or furlough	-0.000	-0.000
	(0.000)	(0.001)
Job loss	0.005	0.004
	(0.004)	(0.003)
Key worker, no employment disruption	-0.000	-0.000
	(0.001)	(0.001)
Self-employed only at baseline	0.001	0.001
	(0.001)	(0.001)
Financial factors		
Baseline financial difficulties	0.009*	0.009*
	(0.005)	(0.005)
Financial situation improved	-0.006	-0.007
	(0.004)	(0.005)
Financial situation worsened	0.008*	0.008*

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	(1) Pooled ref.	(2) WM ref.
	(0.004)	(0.004)
Housing		
Bedroom-per-person ratio	-0.016	-0.025
	(0.018)	(0.020)
Bedroom-per-person ratio squared	0.005	0.010
	(0.012)	(0.012)
Number of other rooms	-0.008	-0.009
	(0.007)	(0.007)
Number of other rooms squared	0.005	0.006
	(0.004)	(0.004)
Health		
Reported COVID-19 symptoms	0.005*	0.005*
	(0.003)	(0.003)
Respiratory condition	-0.000	0.000
	(0.001)	(0.001)
Cardiovascular condition	0.001	0.002
	(0.004)	(0.004)
Endocrine condition	-0.000	0.000
	(0.001)	(0.001)
Arthritis	0.005*	0.005*

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	(1) Pooled ref.	(2) WM ref.
	(0.003)	(0.003)
Other chronic condition	0.001	0.001
	(0.002)	(0.002)
Household composition		
Single-adult household	0.000	0.000
	(0.001)	(0.000)
Number of dependent children	0.000	-0.000
	(0.009)	(0.010)
Number of dependent children squared	0.007	0.008
	(0.006)	(0.007)
Single-adult household × children	0.000	0.000
	(0.003)	(0.003)
Single-adult household × children squared	-0.001	-0.001
	(0.003)	(0.004)
Demographics		
Female	0.001	0.001
	(0.004)	(0.004)
Age 16–29	0.010*	0.011
	(0.006)	(0.007)
Age 30–49	0.005	0.007

Continued on next page

	(1) Pooled ref.	(2) WM ref.
	(0.009)	(0.010)
Age 50–69	0.005	0.004
	(0.005)	(0.005)
Total composition effect	0.044***	0.040***
	(0.014)	(0.015)
Structural effect attributable to:		
Employment factors		
Reduced hours or furlough	0.021	0.021
	(0.029)	(0.029)
Job loss	0.007	0.008
	(0.011)	(0.012)
Key worker, no employment disruption	0.005	0.005
	(0.020)	(0.020)
Self-employed only at baseline	-0.007	-0.007
	(0.010)	(0.010)
Financial factors		
Baseline financial difficulties	-0.008	-0.009
	(0.038)	(0.040)
Financial situation improved	0.044	0.045

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	(1) Pooled ref.	(2) WM ref.
	(0.028)	(0.028)
Financial situation worsened	0.024	0.024
	(0.023)	(0.023)
Housing		
Bedroom-per-person ratio	-0.296	-0.288
	(0.228)	(0.222)
Bedroom-per-person ratio squared	0.081	0.076
	(0.076)	(0.073)
Number of other rooms	-0.298	-0.296
	(0.242)	(0.240)
Number of other rooms squared	0.124	0.123
	(0.094)	(0.093)
Health		
Reported COVID-19 symptoms	0.006	0.006
	(0.018)	(0.019)
Respiratory condition	0.004	0.004
	(0.018)	(0.018)
Cardiovascular condition	0.013	0.012
	(0.014)	(0.013)
Endocrine condition	-0.023	-0.024

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	(1) Pooled ref.	(2) WM ref.
	(0.019)	(0.019)
Arthritis	-0.008	-0.007
	(0.019)	(0.019)
Other chronic condition	-0.011	-0.010
	(0.025)	(0.024)
Household composition		
Single-adult household	-0.012	-0.012
	(0.009)	(0.009)
Number of dependent children	0.018	0.018
	(0.067)	(0.070)
Number of dependent children squared	-0.014	-0.014
	(0.038)	(0.039)
Single-adult household × children	0.010	0.010
	(0.018)	(0.018)
Single-adult household × children squared	-0.002	-0.001
	(0.013)	(0.013)
Demographics		
Female	-0.027	-0.027
	(0.043)	(0.043)
Age 16–29	-0.010	-0.011

Continued on next page

	(1) Pooled ref.	(2) WM ref.
	(0.066)	(0.067)
Age 30–49	-0.043	-0.045
	(0.103)	(0.105)
Age 50–69	-0.014	-0.013
	(0.058)	(0.057)
Constant	0.404	0.404
	(0.325)	(0.325)
Total structural effect	-0.013	-0.009
	(0.045)	(0.046)
Observations	12,533	12,533

Notes: The dependent variable is the individual change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020. The reported gap is defined as White majority (WM) minus BAME; positive values therefore indicate a larger deterioration among BAME respondents relative to the White majority. Column (1) uses pooled coefficients as the reference coefficients, while Column (2) uses White-majority coefficients. The table reports detailed variable-level contributions underlying the aggregate decomposition in Table 8. Standard errors are reported in parentheses. The omitted categories are no change in employment status, age 70 or above, and male; the no-change employment category also includes respondents who were not working before the pandemic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3: Multivariate Regression Estimates for Changes in Mental Well-Being: BIP and White Majority

	(1) Pooled model	(2) BIP	(3) WM
BIP	-0.133**		
	(0.064)		
Employment and financial factors			
Reduced hours or furlough	0.031	0.225*	0.026
	(0.042)	(0.129)	(0.042)
Job loss	-0.179*	-0.220	-0.163*
	(0.094)	(0.254)	(0.096)
Key worker, no employment disruption	0.043	0.119	0.040
	(0.053)	(0.152)	(0.053)
Self-employed only at baseline	-0.062	0.007	-0.063
	(0.052)	(0.183)	(0.052)
Baseline financial difficulties	-0.101**	-0.386***	-0.088*
	(0.047)	(0.129)	(0.047)
Financial situation improved	0.229***	-0.000	0.237***
	(0.047)	(0.180)	(0.047)
Financial situation worsened	-0.224***	-0.633***	-0.207***
	(0.045)	(0.153)	(0.045)
Housing			

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	(1) Pooled model	(2) BIP	(3) WM
Bedroom-per-person ratio	-0.105	-0.146	-0.114
	(0.097)	(0.319)	(0.097)
Bedroom-per-person ratio squared	0.018	0.146*	0.018
	(0.025)	(0.085)	(0.024)
Number of other rooms	-0.044	-0.236	-0.038
	(0.031)	(0.202)	(0.031)
Number of other rooms squared	0.005	0.030	0.004
	(0.003)	(0.038)	(0.003)
Health			
Reported COVID-19 symptoms	-0.089*	-0.144	-0.086*
	(0.049)	(0.201)	(0.049)
Respiratory condition	-0.002	-0.187	0.002
	(0.052)	(0.317)	(0.052)
Cardiovascular condition	0.021	0.099	0.019
	(0.046)	(0.245)	(0.045)
Endocrine condition	-0.012	0.224	-0.018
	(0.048)	(0.139)	(0.048)
Arthritis	0.108**	-0.400	0.115**
	(0.054)	(0.377)	(0.052)

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	(1) Pooled model	(2) BIP	(3) WM
Other chronic condition	0.026	0.245	0.022
	(0.051)	(0.229)	(0.050)
Household composition			
Single-adult household	0.057	0.295	0.041
	(0.140)	(0.267)	(0.139)
Number of dependent children	0.009	0.166	0.002
	(0.050)	(0.160)	(0.051)
Number of dependent children squared	-0.017	-0.038	-0.016
	(0.014)	(0.032)	(0.015)
Single-adult household × children	-0.201	-0.034	-0.187
	(0.312)	(0.769)	(0.306)
Single-adult household × children squared	0.110	0.090	0.107
	(0.127)	(0.563)	(0.124)
Demographics			
Female	-0.188***	0.008	-0.195***
	(0.032)	(0.103)	(0.032)
Age 16–29	-0.117	0.224	-0.127
	(0.078)	(0.278)	(0.078)
Age 30–49	-0.050	0.190	-0.050

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	(1) Pooled model	(2) BIP	(3) WM
	(0.079)	(0.354)	(0.078)
Age 50–69	0.038	0.212	0.039
	(0.053)	(0.254)	(0.052)
Constant	0.109	-0.126	0.111
	(0.111)	(0.413)	(0.111)
Observations	11,385	718	10,667

Notes: The dependent variable is the individual change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020. Higher values indicate better mental well-being; negative coefficients indicate larger deterioration in mental well-being. Column (1) reports estimates from the multivariate pooled specification. Columns (2) and (3) report estimates from group-specific specifications for the minority group and the White majority, respectively. In Column (1), the BIP coefficient is defined relative to the White majority. All models use COVID-19 cross-sectional survey weights. Standard errors, reported in parentheses, account for the survey design. The omitted categories are no change in employment status, age 70 or above, and male; the no-change employment category also includes respondents who were not working before the pandemic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Detailed Oaxaca–Blinder Decomposition of the WM–BIP Gap in Changes in Mental Well-Being

	(1) Pooled ref.	(2) WM ref.
Raw WM–BIP gap	0.187***	0.187***
	(0.060)	(0.060)
Composition effect attributable to:		
Employment factors		
Reduced hours or furlough	0.002	0.002
	(0.003)	(0.003)
Job loss	0.006	0.005
	(0.004)	(0.004)
Key worker, no employment disruption	-0.000	-0.000
	(0.001)	(0.001)
Self-employed only at baseline	-0.001	-0.001
	(0.001)	(0.001)
Financial factors		
Baseline financial difficulties	0.007	0.006
	(0.005)	(0.004)
Financial situation improved	-0.002	-0.002
	(0.007)	(0.008)
Financial situation worsened	0.016**	0.015**

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	(1) Pooled ref.	(2) WM ref.
	(0.007)	(0.006)
Housing		
Bedroom-per-person ratio	-0.038	-0.041
	(0.032)	(0.033)
Bedroom-per-person ratio squared	0.015	0.015
	(0.019)	(0.020)
Number of other rooms	-0.010	-0.009
	(0.007)	(0.007)
Number of other rooms squared	0.005	0.005
	(0.004)	(0.004)
Health		
Reported COVID-19 symptoms	0.001	0.001
	(0.002)	(0.002)
Respiratory condition	-0.000	0.000
	(0.003)	(0.004)
Cardiovascular condition	0.003	0.002
	(0.005)	(0.005)
Endocrine condition	0.000	0.001
	(0.002)	(0.002)
Arthritis	0.008**	0.008**

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	(1) Pooled ref.	(2) WM ref.
	(0.004)	(0.004)
Other chronic condition	0.003	0.003
	(0.005)	(0.005)
Household composition		
Single-adult household	0.001	0.001
	(0.003)	(0.003)
Number of dependent children	-0.002	-0.001
	(0.013)	(0.014)
Number of dependent children squared	0.015	0.014
	(0.012)	(0.013)
Single-adult household × children	-0.008	-0.008
	(0.011)	(0.011)
Single-adult household × children squared	0.010	0.009
	(0.009)	(0.010)
Demographics		
Female	-0.015**	-0.015**
	(0.007)	(0.007)
Age 16–29	0.027	0.030*
	(0.017)	(0.017)
Age 30–49	0.005	0.004

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	(1) Pooled ref.	(2) WM ref.
	(0.006)	(0.007)
Age 50–69	0.007	0.007
	(0.009)	(0.009)
Total composition effect	0.054**	0.053**
	(0.026)	(0.026)
Structural effect attributable to:		
Employment factors		
Reduced hours or furlough	-0.045	-0.045
	(0.029)	(0.029)
Job loss	0.003	0.003
	(0.013)	(0.014)
Key worker, no employment disruption	-0.015	-0.015
	(0.027)	(0.027)
Self-employed only at baseline	-0.004	-0.004
	(0.010)	(0.010)
Financial factors		
Baseline financial difficulties	0.096**	0.096**
	(0.042)	(0.043)
Financial situation improved	0.055	0.055

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	(1) Pooled ref.	(2) WM ref.
	(0.037)	(0.037)
Financial situation worsened	0.097***	0.099***
	(0.035)	(0.035)
Housing		
Bedroom-per-person ratio	0.023	0.027
	(0.270)	(0.265)
Bedroom-per-person ratio squared	-0.107	-0.107
	(0.080)	(0.077)
Number of other rooms	0.367	0.366
	(0.341)	(0.340)
Number of other rooms squared	-0.111	-0.110
	(0.141)	(0.141)
Health		
Reported COVID-19 symptoms	0.007	0.007
	(0.022)	(0.022)
Respiratory condition	0.017	0.017
	(0.023)	(0.022)
Cardiovascular condition	-0.006	-0.005
	(0.014)	(0.014)
Endocrine condition	-0.032	-0.032

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	(1) Pooled ref.	(2) WM ref.
	(0.019)	(0.020)
Arthritis	0.023	0.022
	(0.014)	(0.014)
Other chronic condition	-0.021	-0.020
	(0.025)	(0.025)
Household composition		
Single-adult household	-0.006	-0.006
	(0.007)	(0.007)
Number of dependent children	-0.129	-0.131
	(0.112)	(0.114)
Number of dependent children squared	0.042	0.042
	(0.057)	(0.060)
Single-adult household × children	-0.001	-0.001
	(0.007)	(0.007)
Single-adult household × children squared	0.000	0.000
	(0.006)	(0.006)
Demographics		
Female	-0.091**	-0.090**
	(0.042)	(0.042)
Age 16–29	-0.143	-0.146

Continued on next page

	(1) Pooled ref.	(2) WM ref.
	(0.114)	(0.115)
Age 30–49	-0.093	-0.093
	(0.124)	(0.124)
Age 50–69	-0.031	-0.031
	(0.047)	(0.046)
Constant	0.237	0.237
	(0.376)	(0.376)
Total structural effect	0.133**	0.134**
	(0.062)	(0.062)
Observations	11,385	11,385

Notes: The dependent variable is the individual change in the standardized, seasonally adjusted, and inverted GHQ-12 Likert score between the 2019 baseline and April 2020. The reported gap is defined as White majority (WM) minus BIP; positive values therefore indicate a larger deterioration among BIP respondents relative to the White majority. Column (1) uses pooled coefficients as the reference coefficients, while Column (2) uses White-majority coefficients. The table reports detailed variable-level contributions underlying the aggregate decomposition in Table 9. Standard errors are reported in parentheses. The omitted categories are no change in employment status, age 70 or above, and male; the no-change employment category also includes respondents who were not working before the pandemic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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